



TDMBBO: a novel three-dimensional migration model of biogeography-based optimization (case study: facility planning and benchmark problems)

Mehrdad Kaveh¹ · Mohammad Saadi Mesgari¹ · Diego Martín² · Masoud Kaveh³

Accepted: 4 January 2023 / Published online: 20 January 2023

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023

Abstract

The ability to respond quickly to emergency patients depends on the distribution of ambulance stations. Therefore, the location of these stations is an important issue in urban planning. Health center location-allocation is considered as an NP-hard problem. Due to the long computation time, exact methods will not be effective in solving these problems. On the contrary, many meta-heuristic algorithms have been introduced as promising solutions in various engineering applications. But, it is realized that adaptation of the exploration and exploitation for solving complex optimization problems are challenging tasks. To cope with these challenges, a novel three-dimensional migration model of biogeography-based optimization (TDMBBO) has been introduced to optimize the constrained linear p-median problem. In TDMBBO, nonlinear migration rates based on quadratic, cubic, sinusoidal, and hyperbolic tangent functions have been proposed. In most of the previous migration models, one function for the migration rate has been used. The main disadvantage of these models is that emigration and immigration rates follow a single mathematical function. In the proposed model, for the migration rates of each habitat, a special mathematical model is considered that can apply the appropriate migration rate. The behavior of TDMBBO has been examined on two allocation datasets, IEEE CEC benchmark problems, three random datasets, and two real-world optimization problems. To evaluate the performance of TDMBBO, 31 competitive and state-of-the-art meta-heuristics and five BBO algorithms with different migration models (previous studies) have been used. In allocation datasets, geographic information system has been used to select candidate sites. Parametric and nonparametric tests have also been used to evaluate the performance of algorithms. Overall, TDMBBO yields far better results in many aspects than the other algorithms. The TDMBBO results show a high potential for a location-allocation problems. In IEEE CEC problems, TDMBBO showed rapid convergence compared to other algorithms. The results show the TDMBBO's superiority and this algorithm's capability

in solving real-world optimization problems. In the end, some open problems related to TDMBBO are highlighted encouraging future research in this area.

Keywords Location-allocation · P-median problem · Geographic information system · Improved biogeography-based optimization · And meta-heuristic

1 Introduction

Providing the necessary health services for the injured is one of the most important plans in any country, and the responsibility for these important services has been assigned to the emergency centers [1]. Today, according to people's lifestyles, due to unhealthy diets and difficult living conditions in urban areas, many emergency diseases such as heart disease have dramatically increased [2]. Providing patient care services in a short period of time is one of the most important factors in increasing patient survival. The optimal location of emergency centers is an important factor in reducing the time it takes to reach the accident scene [3]. Location-allocation models play an important role in health and emergency planning and serve as a framework for examining accessibility problems, comparing the quality of previous decisions, and providing alternative solutions for changing existing systems [4–7].

The majority of location models for determining the base station of ambulances are presented in the literature review section [5–13]. Due to the vastness of the search space as well as the complexity of measurement theories, the location allocation of the health center is included as an NP-hard problem [4]. Nowadays, many algebraic modeling languages such as Mosel, GAMS, CPLEX, etc. are used for solving linear and nonlinear problems. There are also many derivative-based algorithms such as steepest descent method, Newton Raphson method, and Quasi-Newton method, etc. for tackling nonlinear optimization problems. However, for highly complex and NP-hard problems, these methods sometimes fail to find optimal or near-optimal solutions [14–16]. Due to the long computation time, exact methods will not be effective in solving these problems. On the contrary, meta-heuristics treat the target model as a black box. They do not need to know the shapes and properties of the models. They can deal with non-smooth, non-differentiable, non-convex, and combinatorial optimization problems [17–20].

Over the past few decades, many meta-heuristic algorithms have been introduced as promising solutions in various engineering applications. With the increasing complexity of problems, the need for new meta-heuristics becomes obvious more than before [21–26]. The reasons for this demand can be summarized into five main motivations: (a) simple structure and concepts; (b) derivation-free mechanisms; (c) local optima avoidance; (d) flexibility; (e) simple and effective implementation. High flexibility and ability to deal with a variety of issues have made these algorithms popular compared to conventional optimization techniques and exact methods [27–31].

Numerous studies have been done on location-allocation problems. Zhang et al. have implemented a multi-objective genetic algorithm (GA) to optimize new

healthcare facilities in Hong Kong [14]. Alkadi et al. used particle swarm optimization (PSO) for maximal covering location (MCLP) and compared their findings with GAMS [15]. The PSO result was better than the other results. Zaffar et al. [16] used a local search algorithm to determine the optimal locations of emergency centers with coverage function and survival rates. Using a hybrid GA, Ko et al. modeled location, allocation, and capacity design in emergency medical centers [3]. As can be seen, in all the above studies, meta-heuristic methods have been used to model location-allocation problems.

1.1 Motivation

Dan Simon introduced biogeography-based optimization (BBO) in 2008 [18]. BBO is a population-based meta-heuristic algorithm using the idea of migration and mutation operation of species for solving complex optimization problems. Unlike other global optimization algorithms, population-based algorithms have more chances of exploring the solution space and thus better chances of escaping from local optima. Through exploration and exploitation, BBO always converges to an optimal or near-optimal solution iteratively in an acceptable time. Specifically, BBO provides feasible solutions that can optimize multi-objective fitness functions [6]. BBO has demonstrated good performance on various constrained and unconstrained test functions. It has also been applied to linear (or nonlinear) real-world optimization problems. For this reason, in this paper, the BBO has been chosen to solve the p-median problem. Then we compare the solution quality with other algorithms to see if further improvement can be made.

After the inception of BBO, a lot of modifications and hybridizations are introduced to enhance its performance. Giri et al. developed a novel migration model for BBO which inherits features of the nearest neighbor of the local best individual to be migrated along with a global best individual of the pool. The experimental results confirm proposed BBO draws clear line of other variants of BBO algorithms in terms of population diversity and establish the accuracy of global optimal solution [32]. An et al. developed non-dominated sorting BBO for the multi-objective flexible job-shop scheduling problem. The statistical results exhibit that proposed BBO has a better performance among other state-of-the-art intelligent algorithms by comparing three benchmark scheduling datasets [33]. Zhang et al. introduced worst opposition learning and random-scaled differential mutation BBO (WRBBO). The experimental results show that WRBBO has better optimization efficiency on benchmark function, clustering optimization, and medical image segmentation than quite a few state-of-the-art BBO variants and other algorithms [34].

Giri et al. proposed an effective rule-based classification technique for the prediction of credit risk using a novel BBO algorithm. The results showed that the proposed rule miner in ten independent runs of tenfold cross validation outperforms all of the aforesaid algorithms in terms of predictive accuracy and coverage [35]. Garg and Deep proposed a new Laplacian BBO which introduces a Laplacian migration operator based on the Laplace crossover of real coded GAs. According to the results, Laplacian BBO is an efficient and reliable algorithm for solving not only the

continuous functions but also real-life problems like camera calibration [36]. Linear migration models of Basic BBO get replaced by different migration models by Ma [37]. A brief literature study on BBO is done by Garg and Deep [38].

Due to the complexity of natural ecosystems, linear-based models are too simple to describe complicated phenomena such as migration [20]. In recent years, analysis of models based on Markov theory has shown how different migration models can affect the performance of BBO [19]. In this field, some researchers have presented different migration models, including linear BBO [18], quadratic BBO [19], sinusoidal BBO [19], generalized sinusoidal BBO [37], and logarithmic BBO [6]. In addition, the No Free Lunch (NFL) theorem shows that all the meta-heuristics execute equally on all optimization problems [39]. Hence, there are still problems that have not been solved, or they can be resolved better by new meta-heuristic algorithms. In this paper, a new migration model of BBO is developed.

1.2 Contributions

In this paper, an improved version of BBO algorithm has been introduced to solve the location-allocation problem. In proposed TDMBBO, nonlinear migration rates and three-dimensional models have been used to increase both the exploitation and exploration of the algorithm. The contributions of this paper are summarized as follows:

- In this paper, a novel three-dimensional migration model of BBO (TDMBBO) are introduced to optimize the p-median problem. In TDMBBO, nonlinear migration rates based on quadratic, cubic, sinusoidal, and hyperbolic tangent functions are proposed. The main goal of the proposed method is to apply a more intelligent migration structure and make the algorithm converge faster to the optimal solution.
- In most previous migration models, one function was used for the migration rate. The main disadvantage of these models is that all habitats are treated the same way, and emigration and immigration rates follow a single mathematical function. In proposed TDMBBO, for the migration rates of each habitat with low, medium, and high habitat suitability index (HSI), a special mathematical model is considered that can apply the appropriate migration rate.
- In TDMBBO, the hyperbolic tangent has a much higher emigration rate than other functions. In this function, the chances of selecting elite habitats to produce new habitats are equal. In an ascending hyperbolic tangent, the convergence rate of the algorithm will be improved. When HSI is high, the selection of an elite habitat reduces the convergence rate of algorithm. In TDMBBO, the immigration rate of the hyperbolic tangent function is lower than other functions and changes slowly. Due to these proposed functions, TDMBBO has higher exploitation and exploration power than other BBO algorithms.
- TDMBBO is applied to several optimization problems: a) Two p-median problems; b) Three maximal covering location problem (random datasets); c) IEEE CEC problems; and d) Two real-world optimization problems.

- 31 competitive meta-heuristic algorithms are used to compare with TDMBBO and the performance of the TDMBBO is compared with previous studies.
- For validation, parametric and nonparametric statistical methods such as Wilcoxon's test, Friedman and Hochberg, standard deviation, convergence curve, and run time of algorithms are used.

1.3 Paper organization

The rest of this paper is organized as follows: Sect. 2 elaborates the problem statement and proposed methodology. Section 3 evaluates the performance of TDMBBO algorithm on some well-known benchmark datasets in comparison with some important meta-heuristics. Finally, in Sect. 4, a summary of the results, conclusions, and future works are provided.

2 Problem statement and proposed methodology

In this section, a novel TDMBBO algorithm are introduced. The main processes of this paper and their relationships are shown in Fig. 1. The behavior of TDMBBO is examined on two allocation datasets, IEEE CEC benchmark problems, three random datasets, and two real-world optimization problems. Thirty-one competitive meta-heuristic algorithms are used to evaluate the performance of TDMBBO. In the real dataset, geographic information systems (GIS) and ordinary weighting averaging (OWA) methods are used to select candidate sites and CPLEX solver is also used to validate meta-heuristics results in allocation datasets. Various nonparametric tests are also used to evaluate the performance of algorithms.

2.1 Basic BBO

BBO is a population-based evolutionary algorithm first proposed by Dan Simon in 2008 [18]. It is based on the theory of island biogeography which deals with the migration, speciation, and extinction of the species in a habitat. The solution in BBO is called "habitat" and habitat is considered a vector of its habitat. In addition, the value of each habitat is defined by the habitat suitability index (HSI). The high value of HSI shows a high fitness function of habitat [36]. The three main operators of BBO are migration, mutation, and elitism. Habitats have their special emigration and immigration rates. The mathematical model of the basic BBO algorithm can be formulated as Eqs. (1) and (2) [4].

$$\mu_j(k) = E \times \left(\frac{k(j)}{N} \right) \quad (1)$$

$$\lambda_j(k) = I \times \left(1 - \frac{k(j)}{N} \right) \quad (2)$$

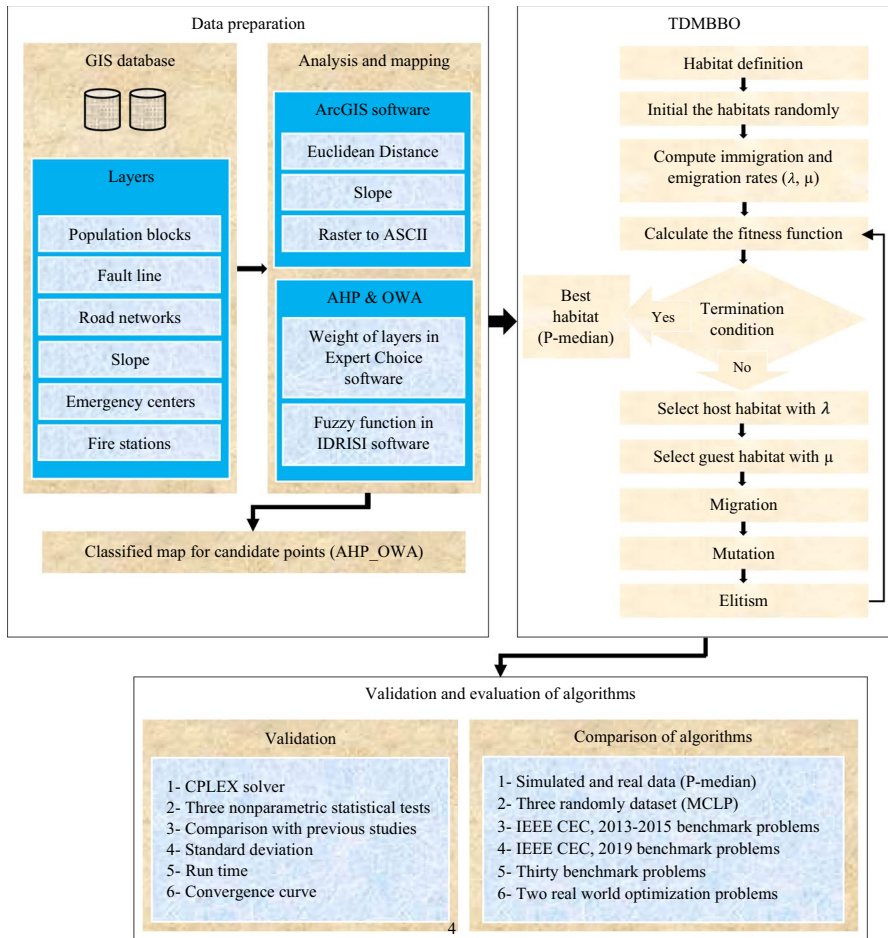


Fig. 1 Flowchart of research method

where $\mu_{k(j)}$ shows emigration rate, $\lambda_{k(j)}$ indicates immigration rate, k_j represents the rank of species, N is population size, E and I display the highest rates of emigration and immigration, respectively. In BBO, the emigration and immigration rates of each solution are used to probabilistically share information between habitats. By assuming H_j as the host habitat and H_i as the guest habitat, the migration process for the standard BBO will be as Eq. (3):

$$H_j(\text{SIVs}) \leftarrow H_j(\text{SIVs}) + H_i(\text{SIVs}) \quad (3)$$

According to Eq. (3), the host habitat (selected based on the immigration rate) receives information from the guest habitat (selected based on the emigration rate) and itself. Also, the mutation operator defines as Eq. (4) [19, 38]:

$$m_j = m_{\max} \times \left(1 - \frac{p_j}{p_{\max}} \right) \quad (4)$$

where $m_{\max}$ is chosen by user, p_j reveals the probability of species count and $p_{\max}$ is the highest value of p_j .

2.2 Improved BBO

Due to the complexity of natural ecosystems, the linear-based models are too simple to describe complicated phenomena such as migration [20]. Various mathematical models of biogeography, create different migration models for BBO. The linear migration model was shown as Eqs. (1) and (2). In recent years, analysis of models based on Markov theory has shown how different migration models can affect the performance of BBO [19]. In this field, some researchers have presented different migration models, including linear BBO [18], quadratic BBO [19], sinusoidal BBO [19], generalized sinusoidal BBO [37], and exponential-logarithmic BBO [6]. Hence, in this paper, a new migration model is developed for the BBO. Equations (5–7) represent the new migration model. In most of the previous migration models, one function for the migration rate has been used. The main disadvantage of these models is that all habitats are treated the same way, and emigration and immigration rates follow a single mathematical function. In the proposed model, for the migration rates of each habitat, a special mathematical model is considered that can apply the appropriate migration rate. The main goal of the proposed method is to apply a more intelligent migration structure and make the algorithm converge faster to the optimal solution.

$$\begin{cases} \mu_{k(j)} = E \times \left(\frac{k(j)}{N} \right)^2 \\ \lambda_{k(j)} = I \times \left(1 - \left(\frac{k(j)}{N} \right)^3 \right) \end{cases} \quad k(j) < N/p \quad (5)$$

$$\begin{cases} \mu_{k(j)} = \frac{E}{2} \times \left(-\cos \left(\frac{k(j)\pi}{N} \right) + 1 \right) \\ \lambda_{k(j)} = \frac{I}{2} \times \left(\cos \left(\frac{k(j)\pi}{N} \right) + 1 \right) \end{cases} \quad N/p \leq k(j) \leq 2N/p \quad (6)$$

$$\begin{cases} \mu_{k(j)} = \frac{E}{2} \times \left(\tanh \left(\frac{k(j)\pi}{N} - \frac{\pi}{6} \right) + 1 \right) \\ \lambda_{k(j)} = \frac{I}{2} \times \left(-\tanh \left(\frac{k(j)\pi}{N} - \frac{\pi}{6} \right) + 1 \right) \end{cases} \quad k(j) > 2N/p \quad (7)$$

where $P \in \{3, 4, 5, 6\}$ is considered for the above equations. The thresholds represent the points at which emigration and immigration rates change their migration functions. In other words, this parameter can be the optimal moment for changing the functions of the migration rates. Figure 2-a shows the emigration rates, and

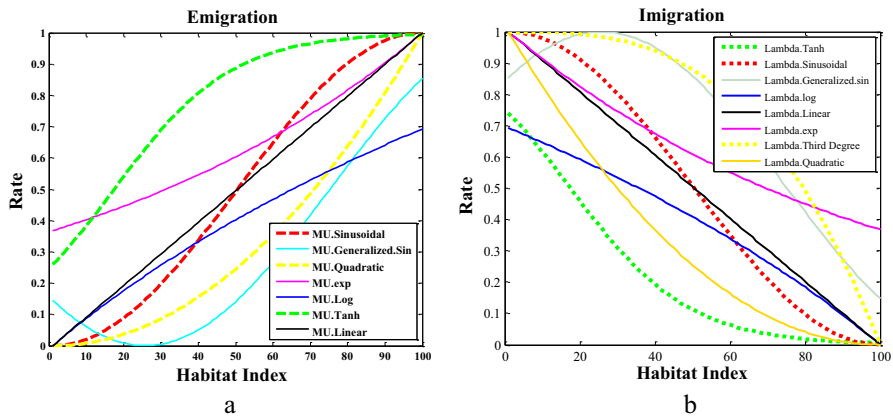


Fig. 2 The curve of mathematical functions for **a** emigration rates, and **b** immigration rates

Fig. 2-b also shows the emigration rates. In the following, the reasons for choosing six functions for the novel TDMBBO have been expressed.

1. As shown in Fig. 2-a, when $\frac{k(j)}{N}$ is small, the emigration rate of the quadratic function is less than the rest and increases slowly from zero. Therefore, in the proposed migration model, when HSI is low, a quadratic function is used. This is because habitats with a lower HSI should have a much lower emigration rate and fewer species that are sent to other habitats. The result of using this logic is a more severe avoidance of weak migrations, which ultimately greatly improves the algorithm's convergence speed.
2. When $\frac{k(j)}{N}$ is moderate, a function is required that the values of its emigration rates are significantly different. Because the probability of choosing a higher HSI is higher than a lower HSI. In Fig. 2-a, the sinusoidal function has the highest slope in the middle areas of the curve, and then the emigration rate of the habitats will be higher.
3. When $\frac{k(j)}{N}$ is large, a function should be used as the emigration model that gives superior values and also changes slowly. In fact, habitats with higher HSI must have higher emigration rates. According to Fig. 2-a, the hyperbolic tangent has a much higher emigration rate than other functions. In this function, the chances of selecting elite habitats to produce new habitats are equal. In an ascending hyperbolic tangent, the convergence rate of the algorithm will be improved.
4. As shown in Fig. 2-b, when $\frac{k(j)}{N}$ is small, we select habitats with lower HSI to have much higher immigration rates. For this reason, a cubic function has been used. The immigration rate of the cubic function is higher than other functions and slowly decreases from its maximum value. The combination of an elite habitat and a very poor habitat always expands the neighborhood's search. The advantage of this model is to eliminate the worry of getting stuck in local minimums and increases the possibility of achieving the best solution.

5. When $\frac{k(j)}{N}$ is moderate, the sinusoidal function is used. For the same reasons for choosing a sinusoidal function to determine the emigration rate, here for the same reasons (and vice versa), the sinusoidal function is used.
6. When $\frac{k(j)}{N}$ is large, we select habitats with higher HSI to have much lower immigration rates. Here, migration can occur in two different cases. The first case is when the chosen habitat through the emigration rate is a habitat with a higher HSI. If a habitat selected through its immigration rate is a habitat with high HSI, we will combine two elite habitats. In this case, it only costs for superior responses. The second case is that the habitat selected through the emigration rate is a habitat with lower HSI, and a habitat selected through its immigration rate is a habitat with higher HSI. In this case, habitats with lower HSIs may migrate to habitats with higher HSIs and possibly destroy the host habitat. Therefore, in both cases, the selection of an elite habitat as the host habitat through its immigration rate is not very useful and even reduces the convergence rate of algorithm. For these reasons, the hyperbolic tangent function is used. The immigration rate of the hyperbolic tangent is lower than other functions and changes slowly.

By choosing the six mentioned mathematical functions for migration rates, the TDMBBO uses various functions appropriate in the combination of habitats with different HSI that will improve its performance. The pseudo-code of TDMBBO

Table 1 Migration models with different mathematical function

Algorithm	Emigration rate	Immigration rate
BBO_linear (basic BBO) [18]	$\mu_{k(j)} = E \times \left(\frac{k(j)}{N} \right)$	$\lambda_{k(j)} = I \times \left(1 - \frac{k(j)}{N} \right)$
BBO_quadratic [19]	$\mu_{k(j)} = E \times \left(\frac{k(j)}{N} \right)^2$	$\lambda_{k(j)} = I \times \left(1 - \frac{k(j)}{N} \right)^2$
BBO_sinusoidal [19]	$\mu_{k(j)} = \frac{E}{2} \times \left(-\cos \left(\frac{k(j)\pi}{N} \right) + 1 \right)$	$\lambda_{k(j)} = \frac{I}{2} \times \left(\cos \left(\frac{k(j)\pi}{N} \right) + 1 \right)$
BBO_generalized sinusoidal [37]	$\mu_{k(j)} = \frac{E}{2} \times \left(-\cos \left(\frac{k(j)\pi}{N} + \varphi \right) + 1 \right)$	$\lambda_{k(j)} = \frac{I}{2} \times \left(\cos \left(\frac{k(j)\pi}{N} + \varphi \right) + 1 \right)$
BBO_exponential logarithmic [6]	$\begin{cases} \mu_{k(j)} = E \times \text{Ln} \left(\frac{k(j)}{N} + 1 \right) & k(j) \leq \frac{N}{3} \\ \mu_{k(j)} = E \times \exp \left(\frac{k(j)}{N} - 1 \right) & k(j) > \frac{N}{3} \end{cases}$	$\begin{cases} \lambda_{k(j)} = I \times \exp \left(-\frac{k(j)}{N} \right) & k(j) \leq \frac{N}{3} \\ \lambda_{k(j)} = I \times \text{Ln} \left(2 - \frac{k(j)}{N} \right) & k(j) > \frac{N}{3} \end{cases}$
TDMBBO [Our] (P=4)	$\begin{cases} \mu_{k(j)} = E \times \left(\frac{k(j)}{N} \right)^2 & k(j) < \frac{N}{4} \\ \mu_{k(j)} = \frac{E}{2} \times \left(-\cos \left(\frac{k(j)\pi}{N} \right) + 1 \right) & \frac{N}{4} \leq k(j) \leq \frac{N}{2} \\ \mu_{k(j)} = \frac{E}{2} \times \left(\tanh \left(\frac{k(j)\pi}{N} - \frac{\pi}{6} \right) + 1 \right) & k(j) > \frac{N}{2} \end{cases}$	$\begin{cases} \lambda_{k(j)} = I \times \left(1 - \left(\frac{k(j)}{N} \right)^3 \right) & k(j) < \frac{N}{4} \\ \lambda_{k(j)} = \frac{I}{2} \times \left(\cos \left(\frac{k(j)\pi}{N} \right) + 1 \right) & \frac{N}{4} \leq k(j) \leq \frac{N}{2} \\ \lambda_{k(j)} = \frac{I}{2} \times \left(-\tanh \left(\frac{k(j)\pi}{N} - \frac{\pi}{6} \right) + 1 \right) & k(j) > \frac{N}{2} \end{cases}$

is shown in Algorithm 1, where t is the current iteration number. Table 1 shows previous migration models and our nonlinear migration model of BBO. The first five rows show the previous models (mentioned in the literature review) and the last row shows our proposed model.

Algorithm 1 Pseudo-codes of TDMBBO

```

Begin
  Set parameters and randomly initialize the population ( $N$ )
  Calculate the HSI of each habitat and sort the population from the best to the worst
  for  $t = 1$  to Max iteration do
    Calculate the immigration and emigration rates by Eq. (5)–(7), calculate mutation rate, and keep some elitist habitats
    for  $j = 1$  to  $N$  do
      for  $k = 1$  to  $D$  (dimension) do
        %% Habitat migration:
        Select habitat  $H_j$  with probability based on immigration rate ( $\lambda_j$ )
        if  $H_j$  is selected Then
          Select habitat  $H_i$  ( $i = 1$  to  $N$ ) with probability based on emigration rate ( $\mu_j$ )
          if  $H_i$  is selected Then
            Select  $k^{th}$  variable of emigrating ( $\mu_j$ ) habitat  $H_i$ 
            Replace selected variable in the immigrating ( $\lambda_j$ ) habitat  $H_j$ 
            Update  $H_j$  (SIV $_k$ ) by Eq. (3)
          end
        end
        %% Habitat mutation:
        if  $rand < m_j$  then
          Select  $k^{th}$  variable of  $H_j$  with probability  $m_j$ 
          if  $H_{jk}$  is selected Then
            Replace  $H_{jk}$  with randomly generated SIV
          end
        end
      end
    end
    Calculate the HSI and sort the population from the best to the worst
  end
End

```

2.3 The p-median problem

The p-median problem was first formulated by Hakimi [7]. The first mathematical model of this problem was presented by ReVelle and Swain [8]. Numerous studies have been performed on the p-median problem [9–12]. As presented, in the majority of these studies, heuristics, and meta-heuristics methods have been used. Kariv and Hakimi have proved the p-median problem is a kind of NP-hard problem on a general network for an arbitrary p [13]. Hence, researchers have used various heuristics and meta-heuristics for this problem. The p-median problem can be formulated using Eqs. (8)–(13) [3].

$$\text{Objective Function} = \text{Minimize} \sum_{i \in I} \sum_{j \in J} D_{ij} Y_{ij} \quad (8)$$

$$\sum_{i \in I} Y_{ij} = 1 \quad \forall j \in J \quad (9)$$

$$Y_{ij} - X_i \leq 0 \quad \forall i \in I, j \in J \quad (10)$$

$$\sum_{i \in I} X_i = P \quad (11)$$

$$Y_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J \quad (12)$$

$$X_i \in \{0, 1\} \quad \forall i \in I \quad (13)$$

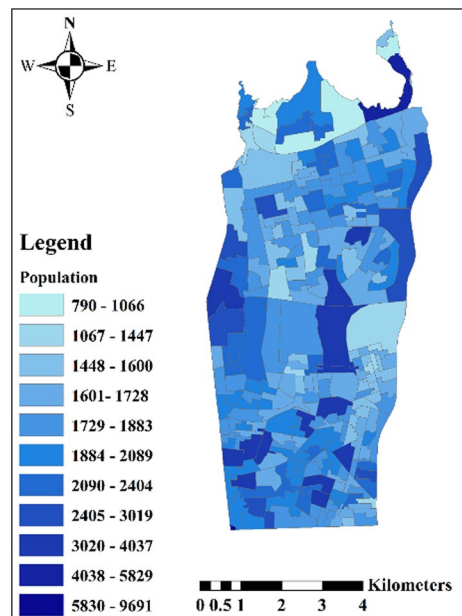
where i , index of facility locations; j , index of demand points; D_{ij} , the shortest distance from i to j ; P , number of facilities;

$$X_i = \begin{cases} 1, & \text{if a service center should be located at the place } i, \\ 0, & \text{otherwise.} \end{cases}$$

$$Y_{ij} = \begin{cases} 1, & \text{if assign the users location } j \text{ to the possible location } i, \\ 0, & \text{otherwise.} \end{cases}$$

Constraints (9) ensure that each demand is assigned to exactly one possible service center. Constraints (10) prevent any demand from being supplied from a site with an unopened center, and constraint (11) limits the number of facilities.

Fig. 3 Population density map of study area



2.3.1 Real dataset

In this paper, the spatial data set of Region 2 of Tehran has been used. Figure 3 indicates the spatial distribution of the population in this area. According to the 2011 statistical yearbook of Iran, the population density in this region is about 624 thousand people in the whole region. Currently, this area has two emergency centers that are not well distributed due to the high population density. ArcGIS 10.4.4 software was used to generate distance maps using "Euclidean distance" and "slope" analysis. Figure 4 shows the criteria maps. As can be seen, six criteria have been created: distance from existing emergency bases, distance from fire stations, distance from population centers, distance from the road network, distance from the fault, and the slope of the region.

In this paper, the values of the weights for layers have been calculated using the analytic hierarchy process (AHP) in Expert Choice software. The AHP method is one of the most extensively used and popular multi-criteria decision-making techniques [40]. The processes of assessing alternatives and aggregating them to determine the most relevant alternatives are included in this technique. The technique

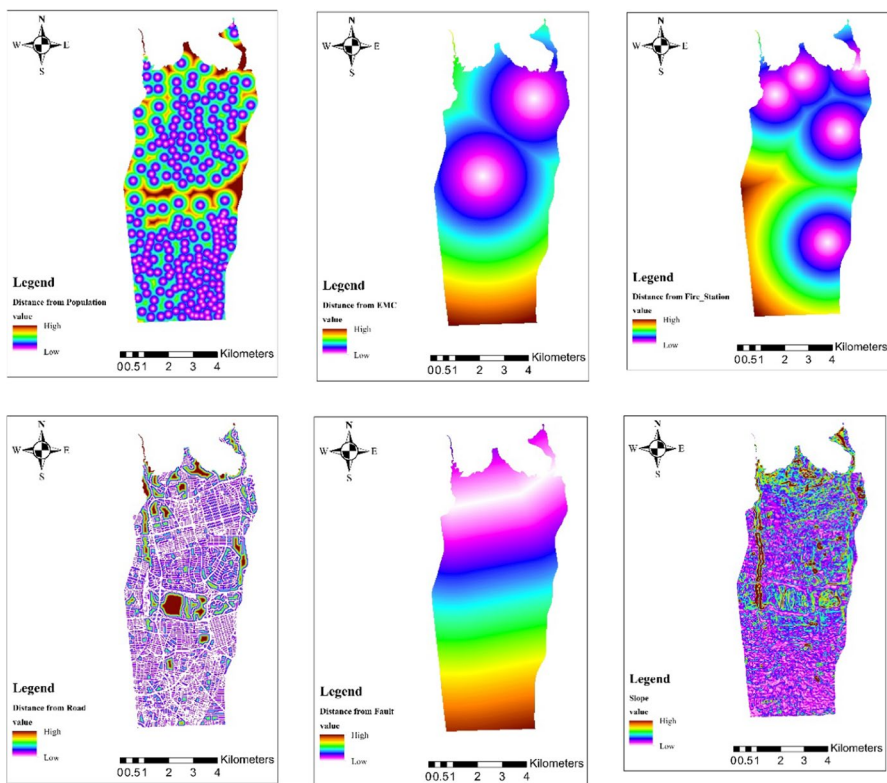
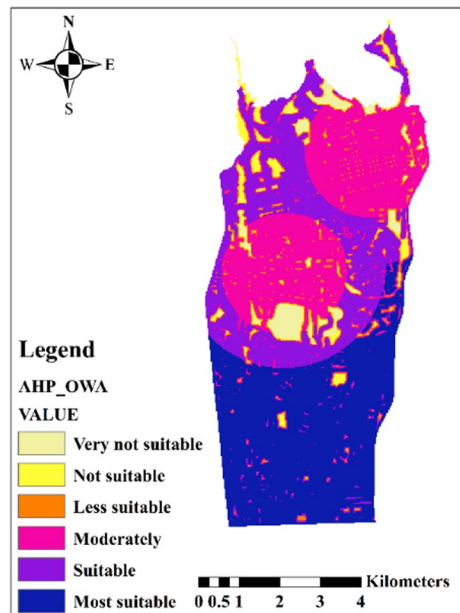


Fig. 4 Criteria maps

Table 2 The threshold and fuzzy function type of the criteria maps

Layers	Threshold				Fuzzy function type	Fuzzy function shape
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>		
Population blocks	–	–	2000	3500	Linear	Monotonically-decreasing
Fault line	2500	8000	–	–	Linear	Monotonically-increasing
Road networks	–	–	35	100	Linear	Monotonically-decreasing
Slope	–	–	25	40	Linear	Monotonically-decreasing
Emergency centers	1000	2500	6000	10,000	Linear	Symmetric
Fire stations	–	–	3000	7000	Linear	Monotonically-decreasing

Fig. 5 Classified map for locating emergency centers

is used to rank a set of options or to select the best choice from a set of alternatives. ASCII maps were generated using six benchmark maps based on "Raster to ASCII" analysis. Finally, the fuzzy functions of each criterion have been defined in the IDRISI software. The other factors that influence the standardization of fuzzy maps are the thresholds, which are referred to as control points. Table 2 shows the threshold and fuzzy function types of the criteria maps. Figure 5 shows the output map of the geographic information system (GIS), the AHP, and the ordinary weighting averaging (OWA) process. To find the candidate points, three classes were used: the most suitable, moderately suitable, and medium suitable.

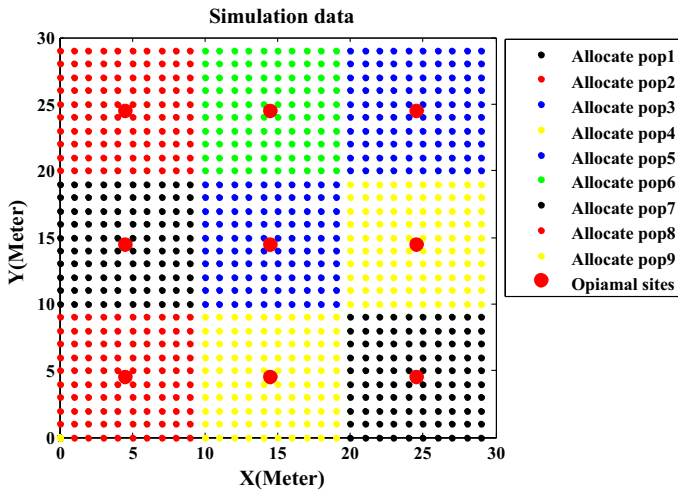


Fig. 6 Simulated data

2.3.2 Simulated dataset

In this paper, a set of regular simulated data is used (Fig. 6). These data include up to 900 blocks that are distributed regularly over a network with a compatible distance. The aim is to find nine centers and allocate all the blocks to the closest centers so that the total distance between the blocks and the centers related to them reaches the shortest possible distance. If the district is divided into nine equal parts with equal shape and area, the centers of gravity of these parts will be the best locations. If the algorithm works well, it should be able to identify these nine centers as the desired solutions. Thus, these data can be applied for the evaluation and validation of the algorithms.

2.3.3 Problem formulation in TDMBBO

In this paper, the number of species in each habitat is shown as the number of centers. Nine emergency centers are also required to create emergency facilities using the set covering model. Figure 7 shows the definition of the habitat.

The roulette wheel method has been used to select the host and guest's habitats using the emigration rate and immigration rate. Then to apply the migration process, a random number of the habitant from the guest's habitat to the host's habitat is sent. Figure 8 shows an example of migration for TDMBBO.

Also, Fig. 9 shows an example of the mutation process and a number of habitats enter the next generation as elite solutions.

Habitat	200	1001	82	451	770	901	500	881	139
---------	-----	------	----	-----	-----	-----	-----	-----	-----

Fig. 7 Encoding an solution in form of a habitat

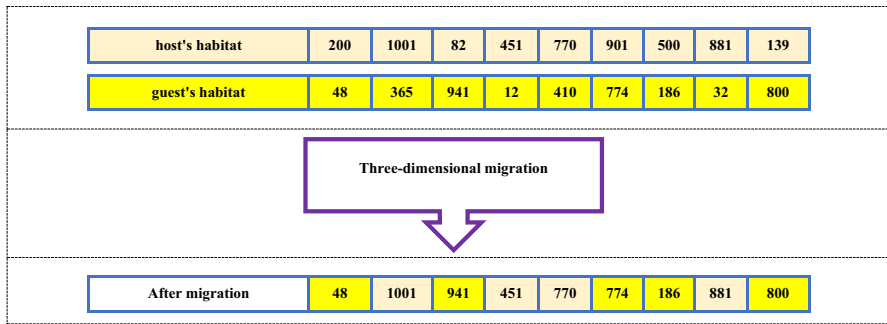


Fig. 8 Example of migration process

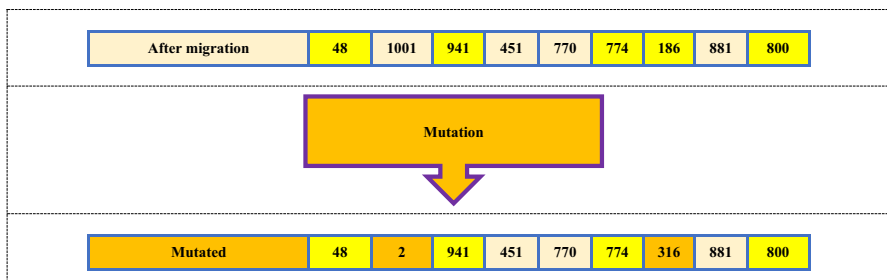


Fig. 9 An example of a mutation process

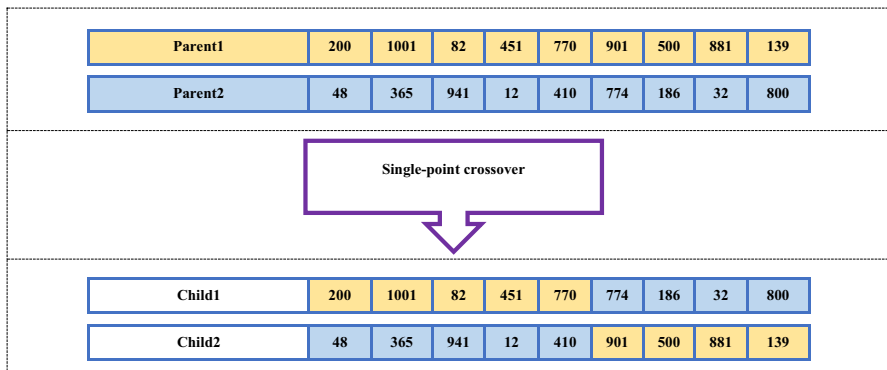


Fig. 10 An example of single-point crossover

2.3.4 Problem formulation in the GA

The definition of chromosomes in the GA is similar to the definition of habitat in the TDMBBO. Numerous chromosomes are randomly generated to create the

initial population. The roulette wheel approach is used to choose a parent. Each generation's parents are chosen using a roulette wheel. A chromosome with a higher fitness function has a greater likelihood of selection. The next step, after selecting the parents, is to first check the probability of performing a crossover operation based on the crossover rate. Then, according to Fig. 10, a single-point crossover operation is performed. Also, Fig. 9 shows an example of the mutation process, in which a number of chromosomes enter the next generation as elite solutions.

2.3.5 Problem formulation in PSO

The particles in PSO have a definition similar to that of habitat in the TDMBBO (Fig. 7). After initial population generation, the particles are introduced as their best personal experience (Pbest). As well, particles with the best fitness function have the best global experience (Gbest) [41, 42]. According to the problem in this paper, which is discrete, the mutation operation of TDMBBO has been used to model the motion of inertia. Since the previous motion is a random section of the PSO. To move toward the best personal experience, some decision parameters of the best personal experience are randomly replaced in the particle. To model the movement toward the best global experience, the best particles of each generation are used so that this movement does not spoil the effect of the previous movement. Figure 11 illustrates an example of these movement.

3 Results and discussion

In this section, the performance of the TDMBBO is examined in seven-parts: Two p-median problems; a maximal covering location problem (three random datasets); IEEE CEC benchmark problems; Comparison with previous studies; and two real world optimization problems.

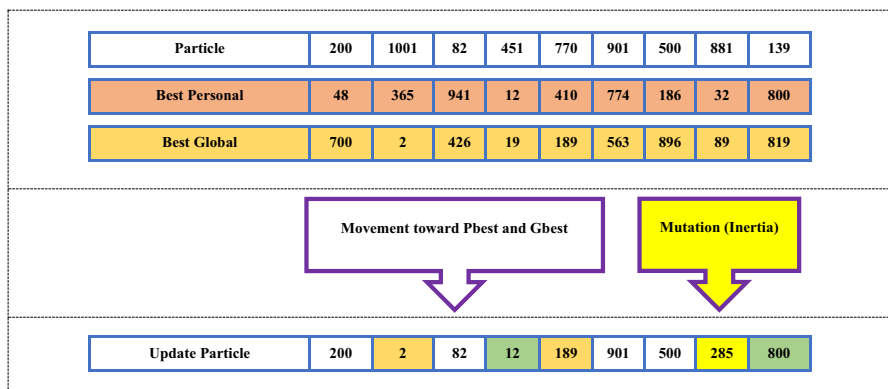


Fig. 11 Example of movement toward the Inertia, Pbest, and Gbest

Meta-heuristic algorithms have several parameters that should be calibrated. Therefore, the best combination these parameters must be determined before evaluating the algorithm. In this paper, the simulated datasets are used to adjust the TDMBBO and the calibration of TDMBBO parameters is set by trial and error method. Since the optimal solutions for the simulated datasets are recognized, one can obtain the best combination of parameters. The best fitness function value in the simulated datasets are 3430.7527 m. For finding the best value for each of the parameters, other variables were kept fixed and the algorithm was implemented with different values of parameters. Also, the value of fitness function is considered as the main criterion for the measurement and calibration of the algorithm's parameters. This dataset is a minimization problem. Therefore, the lower values of the fitness function mean better parameters settings. In this method, for each calibration parameter, have been selected and tested a large number of values. But briefly, and to show them in the table, have been considered only a limited number of instance and have been presented their results. Table 3 shows the results of TDMBBO using various parameters.

According to Table 3, the mutation rate of 0.15 produced the best results. The larger values lead to less convergence. The low level of elitism, including 3%, reduces the convergence rate of the algorithm. The best elitism rate is 10%. The threshold (p) represents the points at which emigration and immigration rates change their migration functions. The best value of this parameter is four. With the initial population of 50 habitats, the algorithm did not find the appropriate solution. By increasing populations to 150 habitats, the algorithm has been improved, but the optimal solution has not been discovered. Finally, with 200 habitats, the algorithm has been able to determine the optimal solution. In the iteration of 100, the algorithm has been far from the optimal solution. In the iteration of 200, the algorithm has been approached the optimal solution and in the iteration of 300, the algorithm has been found the optimal solution. The best-found values for the TDMBBO (and other meta-heuristic algorithms) parameters are presented in Table 4.

3.1 P-median problem

To evaluate the performance of TDMBBO, two well-known algorithms called PSO, and GA, and six state-of-the-art competitive algorithms called iterated greedy (IG) [11], improved crow search algorithm (I-CSA) [12], general variable neighborhood search (GVNS) [43], capuchin search algorithm (CapSA) [44], chimp optimization algorithm (ChOA) [45], and grey wolf optimizer (GWO) [46], and five BBO algorithms with different migration models have been used.

Table 5 shows the results of the algorithms in both allocation datasets. According to Table 4, TDMBBO has provided the best results. The best fitness function value in the real dataset is 172627.7840 m. Centers with numbers {105, 145, 355, 411, 597, 606, 913, 954, and 994} are the optimal centers found by the TDMBBO. TDMBBO has the best standard division in both datasets, and the average runtime is less than the others. I-CSA and CapSA also performed well. Figure 12 indicates the

Table 3 Calibration of the TDNBBO parameters using trial and error method

No. of run	Mutation rate	Elitism percent	Threshold ($p \in \{3, 4, 5, 6\}$)	Migrating rate	Population size	Iteration	Fitness function (m)
1	0.15	10%	4	[0, 1]	200	300	3430.7527
2	0.35	10%	4	[0, 1]	200	300	3468.1296
3	0.04	10%	4	[0, 1]	200	300	3472.1296
4	0.15	3%	4	[0, 1]	200	300	3459.1296
5	0.15	30%	4	[0, 1]	200	300	3463.3962
6	0.15	60%	4	[0, 1]	200	300	3476.1950
7	0.15	10%	3	[0, 1]	200	300	3453.9652
8	0.15	10%	5	[0, 1]	200	300	3458.9526
9	0.15	10%	6	[0, 1]	200	300	3462.9617
10	0.15	10%	4	[0, 0.8]	200	300	3452.9625
11	0.15	10%	4	[0, 0.5]	200	300	3475.9563
12	0.15	10%	4	[0, 0.2]	200	300	3492.8516
13	0.15	10%	4	[0, 1]	50	300	3470.7425
14	0.15	10%	4	[0, 1]	100	300	3435.1475
15	0.15	10%	4	[0, 1]	150	300	3431.2856
16	0.15	10%	4	[0, 1]	200	100	3435.9625
17	0.15	10%	4	[0, 1]	200	200	3432.1935
18	0.15	10%	4	[0, 1]	200	300	3430.7527

Table 4 Calibration of algorithm parameters by trial and error method (Population size=200, Iteration=300)

Algorithm	Parameter	Value
TDMBBO	The probability range for migrating into for each gene	[0, 1]
	Threshold (P)	4
	Elitism percent	10%
	Mutation rate	0.15
IG	T	0.40
	d	4
I-CSA	Flight length (fl)	2
	Awareness probability (AP)	0.10
GVNS	Number of Neighborhoods ($K_{\max}$)	3
	Number of Shakings	25
ChOA	a	$[-1, 1]$
	f	Linearly decreased from 2 to 0
CapSA	Velocity control constants	1.00
	Inertia parameter	0.68
	Balance and elasticity factors	0.70, 9
GWO	C	0.7
	A	0.3
	α	Linearly decreased from 2 to 0
ACO	β	2
	ρ, α	0.10
	q_0	0.90
	Scaling factor (F)	0.85
DE	Crossover probability constant (CR)	0.95
	Mutation rate	0.05
	Elitism percent	16%
GA	Mutation rate	0.20
	Crossover rate	0.84
PSO	The inertial movement rate (α)	0.12
	The movement toward the best personal experience rate ($\Phi 1$)	0.68
	The movement toward the best global experience rate ($\Phi 2$)	0.91

optimal combination of the centers and the allocation of population blocks to them. As can be seen, nine centers were found next to the previous two centers.

To validate the results obtained from meta-heuristics, the CPLEX solver in GAMS software has been utilized. According to the GAMS results, the best value of the fitness function (p-median) was 172,628 m. Also, centers with ID numbers 105, 145, 355, 411, 597, 606, 913, 954, and 994 were elected by the CPLEX. These results were consistent with the results obtained by TDMBBO. Figure 13 shows the average fitness function of all search agents in each iteration.

Table 5 The results of the algorithms for p-median problem

Algorithm	Simulated dataset			Real dataset		
	Average fitness (m)	Average runtime (s)	Standard division	Average fitness (m)	Average runtime (s)	Standard division
TDMBO	3430.7527	22	0.000000	172,627.7840	35	0.000007
IG	3652.1424	38	0.000036	173,026.4561	59	0.000836
I-CSA	3430.7527	32	0.000000	172,814.1863	42	0.000306
GVNS	3725.3214	45	0.000152	173,429.3261	65	0.001028
CapSA	3430.7527	41	0.000000	172,928.1759	51	0.000493
GWO	3725.4369	47	0.000452	173,551.1426	62	0.004468
ChOA	3698.1453	39	0.000263	173,149.1796	59	0.002307
GA	3949.2189	46	0.000963	175,148.1782	64	0.081267
PSO	3902.8745	52	0.000856	174,957.7129	72	0.062574
BBO-linear	3896.3654	36	0.000743	174,619.2185	44	0.052145
BBO-quadratic	3964.1298	39	0.000789	174,329.2596	58	0.040254
BBO-sinusoidal	3878.8965	31	0.000475	174,037.3147	69	0.039741
BBO-generalized sinusoidal	3820.7025	37	0.000502	173,885.1452	72	0.021459
BBO-logarithmic	3810.9628	35	0.000319	173,434.3569	53	0.020145

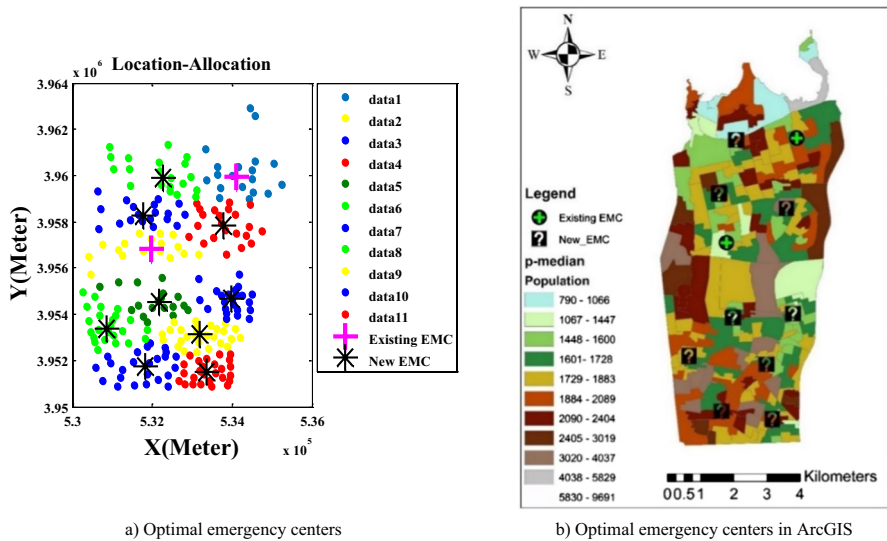


Fig. 12 The optimal emergency centers in study area (real dataset)

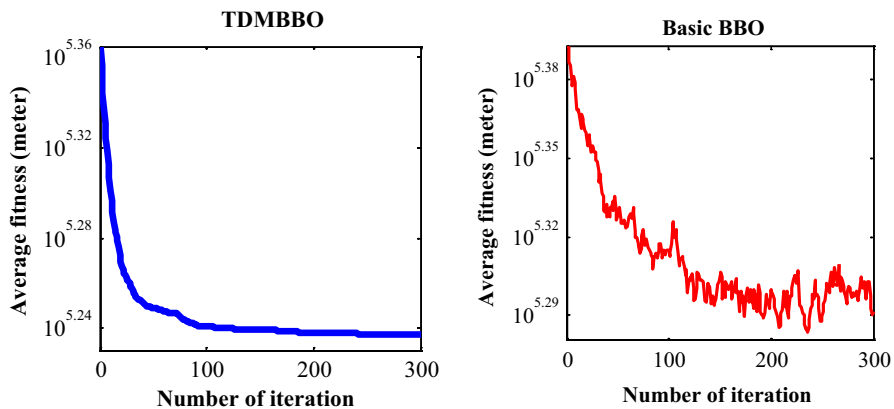


Fig. 13 The average fitness function of all search agents in each iteration

In TDMBBO, all weak habitats move towards improvement by using new migration functions. In basic BBO, weak habitats have little chance to improve. Figure 14 indicates the convergence curve of algorithms. The convergence curve of the TDMBBO is steeper than that of other algorithms. Since the migration in TDMBBO is such that a host habitat sends its habitants to a guest habitat, the fitness function of all habitats is improved.

According to Fig. 14, the TDMBBO converged earlier than other BBO algorithms. In fact, in the proposed TDMBBO, for emigration and immigration rates, a particular mathematical model is utilized so that the algorithm can apply appropriate migration rates for different solutions.

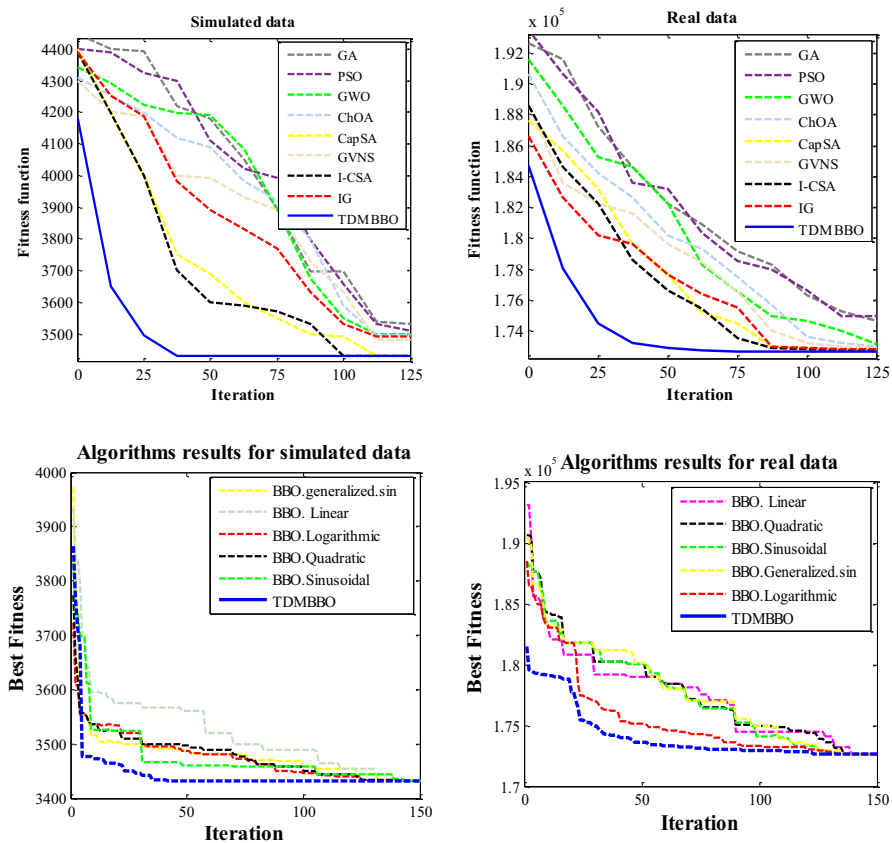


Fig. 14 The convergence trend of algorithms on p-median problem

To evaluate the repeatability of the algorithms, they are executed 40 times on both simulated and real data. Figure 15 shows the best solutions after 40 implementations of TDMBBO and BBO algorithms with previous migration models (linear, quadratic, sinusoidal, generic sinusoidal, logarithmic) on real data. In this figure, the distribution of the fitness function (p-median problem) of algorithms is presented. As can be clearly seen, the TDMBBO algorithm has shown greater repeatability. All algorithms (linear, quadratic, sinusoidal, generic sinusoidal, and logarithmic) have achieved an optimal solution in 80% of the cases, but TDMBBO has found the best solution in over 90% of the cases. According to Table 5, the average runtime of algorithms shows that the runtime of TDMBBO is less than other algorithms. The runtime of all models of the BBO is appropriate. But TDMBBO has a shorter runtime to find the optimal solution compared to its previous models.

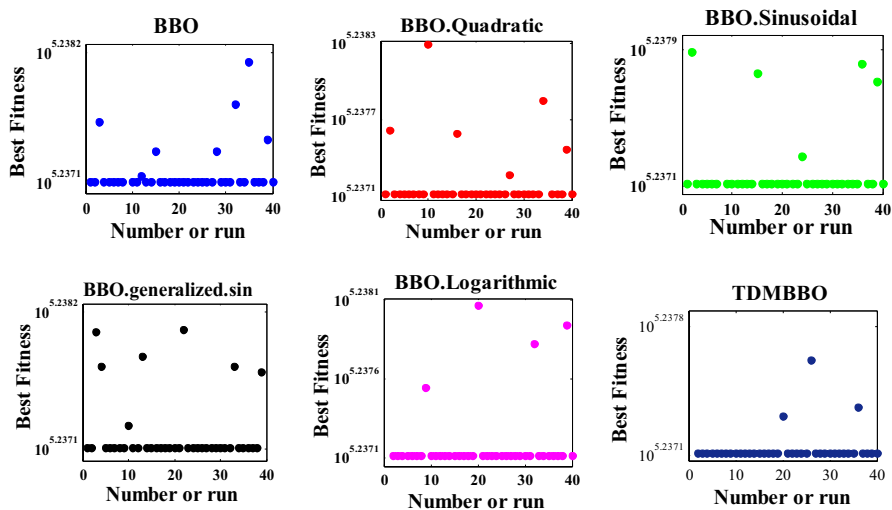


Fig. 15 Repeatability of previous migration models and new migration model in the BBO

Table 6 Three random dataset

Type of data set	Number of dataset	Number of facilities	Number of blocks	Number of population
Small instance	10	1000	300	100 < pop < 1000
Medium instance	10	5000	800	100 < pop < 1000
Large instance	10	10,000	1500	100 < pop < 1000

3.2 Maximum coverage location problem

In this part, three random generate instances have been used. The MCLP emerged in the late 1970s, motivated by the necessity of designing better models to address the position of facilities. Full details of the MCLP model are available in Kaveh and Mesgari [4]. Table 6 shows the details of the randomly generated instances.

Table 7 shows the results of the TDMBBO and other algorithms on small instances. The optimal solutions in all instances have been calculated by CPLEX in GAMS software. In per 10 small random instances, TDMBBO, like CPLEX, obtained the optimal solution. In addition, IG, I-CSA, GVNS, CapCA, ChOA, GWO, PSO, and GA achieved optimal results in 8, 9, 8, 8, 7, 7, 6, and 6 instances, respectively.

Table 8 indicates the results of all algorithms on medium instances. In these instances, CPLEX, TDMBBO, IG, I-CSA, GVNS, CapCA, ChOA, GWO, PSO, and GA obtained the optimal solution in 10, 10, 7, 8, 6, 7, 6, 6, 5, and 5 instances, respectively. As shown in Table 8, CPLEX and TDMBBO are 100% successful.

Table 7 The results of the meta-heuristics in small randomly dataset

Small dataset	Results of meta-heuristics										CPLEX Optimal sol
	TDMBBO	IG	I-CSA	GVNS	CapCA	ChOA	GWO	PSO	GA		
DB1	448,775	448,775	448,775	448,775	448,775	446,874	448,775	448,775	445,159	448,775	
DB2	425,908	425,908	424,265	425,908	425,908	425,908	425,908	424,856	425,908	425,908	
DB3	431,921	430,365	431,921	431,921	431,921	431,921	430,856	431,921	430,193	431,921	
DB4	428,105	428,105	428,105	426,896	427,802	428,105	428,105	426,856	425,698	428,105	
DB5	462,620	462,620	462,620	462,620	462,620	460,574	462,620	462,620	462,620	462,620	
DB6	457,137	457,137	457,137	457,137	455,874	457,137	457,137	457,137	457,137	457,137	
DB7	466,445	464,853	466,445	466,445	466,445	466,445	466,445	464,112	463,834	466,445	
DB8	459,671	459,671	459,671	458,236	459,671	459,671	457,856	459,671	459,671	459,671	
DB9	464,973	464,973	464,973	464,973	464,973	463,547	464,973	463,856	464,973	464,973	
DB10	460,450	460,450	460,450	460,450	460,450	460,450	459,852	460,450	460,450	460,450	

Table 8 The results of the meta-heuristics in medium randomly dataset

Medium dataset	Results of meta-heuristics									CPLEX Optimal sol
	TDMBBO	IG	I-CSA	GVNS	CapCA	ChOA	GWO	PSO	GA	
DB1	1,155,423	1,151,698	1,155,423	1,155,423	1,155,423	1,155,423	1,141,865	1,150,546	1,155,423	1,155,423
DB2	1,167,197	1,167,197	1,167,197	1,162,861	1,167,197	1,167,197	1,167,197	1,167,197	1,167,197	1,167,197
DB3	1,171,982	1,171,982	1,171,982	1,167,183	1,167,816	1,168,647	1,165,186	1,171,982	1,162,219	1,171,982
DB4	1,161,774	1,161,774	1,158,453	1,161,774	1,161,774	1,161,774	1,161,774	1,155,785	1,161,774	1,161,774
DB5	1,154,430	1,150,832	1,154,430	1,154,430	1,150,469	1,149,256	1,151,865	1,150,145	1,148,236	1,154,430
DB6	1,119,571	1,119,571	1,119,571	1,114,826	1,119,571	1,115,369	1,119,571	1,119,571	1,119,571	1,119,571
DB7	1,156,593	1,149,297	1,156,593	1,156,593	1,156,593	1,156,593	1,149,423	1,147,659	1,149,745	1,156,593
DB8	1,109,765	1,109,765	1,101,854	1,109,765	1,109,765	1,109,765	1,109,765	1,109,765	1,109,765	1,109,765
DB9	1,156,123	1,156,123	1,156,123	1,151,853	1,156,123	1,150,736	1,156,123	1,149,265	1,152,854	1,156,123
DB10	1,143,986	1,143,986	1,143,986	1,143,986	1,139,456	1,143,986	1,143,986	1,143,986	1,126,874	1,143,986

Table 9 The results of the meta-heuristics in large randomly dataset

Large dataset	Results of meta-heuristics for MCLP objective function									CPLEX Optimal sol
	TDMBBO	IG	I-CSA	GVNS	CapCA	ChOA	GWO	PSO	GA	
DB1	2,245,698	2,226,852	2,245,698	2,236,149	2,236,514	2,236,215	2,245,698	2,226,984	2,236,548	—
DB2	2,131,842	2,145,645	2,129,845	2,129,214	2,145,645	2,145,645	2,136,254	2,131,656	2,129,325	—
DB3	2,112,365	2,102,843	2,106,845	2,112,365	2,103,269	2,103,265	2,112,365	2,112,365	2,110,236	—
DB4	2,102,364	2,102,364	2,102,364	2,102,364	2,102,364	2,102,364	2,065,985	2,071,659	2,102,364	—
DB5	2,278,314	2,278,314	2,259,632	2,253,269	2,265,269	2,247,985	2,251,236	2,251,326	2,269,412	—
DB6	2,204,565	2,186,169	2,186,369	2,183,218	2,204,565	2,183,265	2,200,215	2,201,369	2,193,265	—
DB7	2,135,269	2,129,265	2,150,325	2,134,128	2,150,325	2,150,325	2,130,389	2,150,325	2,136,985	—
DB8	2,212,378	2,212,378	2,212,378	2,212,378	2,196,325	2,201,362	2,212,378	2,206,548	2,212,378	—
DB9	2,198,654	2,176,183	2,176,286	2,173,429	2,198,654	2,171,256	2,198,654	2,198,654	2,159,652	—
DB10	2,101,245	2,101,245	2,095,845	2,101,245	2,101,245	2,101,245	2,069,269	2,079,652	2,068,451	—

Finally, Table 9 shows the implementation of algorithms upon large instances. In these instances, TDMBBO, IG, I-CSA, GVNS, CapCA, ChOA, GWO, PSO, and GA obtained the optimal solution in 8, 5, 5, 4, 6, 4, 4, 3, and 2 instances, respectively. TDMBBO has a high capability in all three types of random instances. Whereas CPLEX failed to determine the global optimal in large random instances.

3.3 Nonparametric statistical test

In this part, a nonparametric test called Wilcoxon has been used to evaluate the performance of algorithms. The Wilcoxon test was applied to measure the similarity of two dependent degree-scale samples. Derrac et al. [47] provided the full details of this test. All algorithms have been implemented with 25 runs on 10 small instances (Sect. 3.2). The mean values of the fitness function were normalized, and then the Wilcoxon test results were obtained using SPSS software. Table 10 shows the R^+ , R^- , and p -values for all TDMBBO pairwise comparisons. As shown in Table 9, TDMBBO shows an improvement versus IG, I-CSA, GVNS, and CapCA with a level of significance $\alpha=0.05$, and versus ChOA, GWO, PSO, and GA with a level of significance $\alpha=0.01$. According to the results, TDMBBO has a strong performance when compared to other algorithms.

3.4 IEEE CEC benchmark problems

In this part, CEC2005 test functions [48] are used to evaluate the performance of TDMBBO. These functions are rotated, shifted, expanded, and combined with conventional test functions that offer the highest complexity among conventional test functions. Table 11 shows the details of the CEC2005 functions. The 3D plot of these functions is shown in Fig. 24. To evaluate the performance of TDMBBO, three competitive algorithms called GWO, ant colony optimization (ACO), and differential evolution (DE), and five BBO algorithms with different migration models (previous studies) called BBO-logarithmic, BBO-generalized sinusoidal, BBO-sinusoidal, BBO-quadratic, and BBO-linear have been used.

Table 10 The results of algorithms in Wilcoxon test

Comparison of algorithm	R^+	R^-	p -value	Level of significance (α)
TDMBBO versus IG	36	19	0.051	$\alpha=0.05$
TDMBBO versus I-CSA	34	21	0.071	$\alpha=0.05$
TDMBBO versus GVNS	37	18	0.045	$\alpha=0.05$
TDMBBO versus CapCA	35	20	0.062	$\alpha=0.05$
TDMBBO versus ChOA	39	16	0.011	$\alpha=0.01$
TDMBBO versus GWO	41	14	0.009	$\alpha=0.01$
TDMBBO versus PSO	45	10	0.004	$\alpha=0.01$
TDMBBO versus GA	48	7	0.003	$\alpha=0.01$

Table 11 The details of the CEC2005 test functions

No.	Test functions	Range	Dimension	F (x*)
<i>Unimodal functions</i>				
F1	Shifted sphere function	[-100,100]	5	-450
F2	Shifted schwefel's	[-100,100]	5	-450
F3	Shifted rotated high conditioned elliptic function (F4)	[-100,100]	5	-450
F4	Shifted schwefel's with noise in fitness	[-100,100]	5	-450
F5	Schwefel's with global optimum on bounds	[-100,100]	5	-310
<i>Multimodal functions</i>				
F6	Shifted rosenbrock's function	[-100,100]	5	390
F7	Shifted rotated griewank's function without bounds	[0,600]	5	-180
F8	Shifted rotated ackley's function with global optimum on bounds	[-32,32]	5	-140
F9	Shifted rastrigin's function	[-5,5]	5	-330
F10	Shifted rotated rastrigin's function	[-5,5]	5	-330
F11	Shifted rotated weierstrass function	[-0.5,0.5]	5	90
F12	Schwefel's	[-100,100]	5	-460
F13	Expanded extended griewank's plus rosenbrock's function (F8F2)	[-3,1]	5	-130
F14	Shifted rotated expanded scaffer's F6	[-100,100]	5	-300
<i>Composite functions</i>				
F15	Hybrid composition function	[-5,5]	10	120
F16	Rotated hybrid composition function	[-5,5]	10	120
F17	Rotated hybrid composition function with noise in fitness	[-5,5]	10	120
F18	Rotated hybrid composition function	[-5,5]	10	10
F19	Rotated hybrid composition function with a narrow basin for the global optimum	[-5,5]	10	10
F20	Rotated hybrid composition function with the global optimum on the bounds	[-5,5]	10	10

Table 12 The results of unimodal CEC2005 test functions

Algorithm	Metric	CEC _{F1}	CEC _{F2}	CEC _{F3}	CEC _{F4}	CEC _{F5}
TDMBBO	Ave	− 439.16	− 437.26	− 439.73	− 437.16	− 309.26
	Std	5.12	6.69	3.06	4.36	0.15
	<i>p</i> -value	N/A	N/A	N/A	N/A	N/A
GWO	Ave	− 437.36	− 435.18	− 437.36	− 434.18	− 307.39
	Std	12.26	18.51	12.36	18.36	5.29
	<i>p</i> -value	0.0043	0.0063	0.0076	0.2119	0.0018
BBO-logarithmic	Ave	− 436.96	− 435.96	− 430.39	− 433.85	− 308.39
	Std	14.83	15.35	33.14	23.96	3.81
	<i>p</i> -value	0.0385	0.0052	0.4569	0.2259	0.0008
ACO	Ave	− 435.48	− 434.19	− 433.16	− 431.18	− 305.42
	Std	19.39	22.19	24.36	31.82	29.18
	<i>p</i> -value	0.2163	0.0615	0.1635	0.3352	0.1485
BBO-generalized sinusoidal	Ave	− 434.59	− 430.18	− 415.32	− 425.96	− 302.33
	Std	25.68	32.79	43.29	45.29	45.62
	<i>p</i> -value	0.3625	0.4139	0.7536	0.5129	0.3524
BBO-sinusoidal	Ave	− 432.65	− 428.09	− 413.28	− 420.82	− 298.29
	Std	33.49	36.14	48.36	52.36	55.39
	<i>p</i> -value	0.4756	0.5296	0.8136	0.6238	0.5214
BBO-quadratic	Ave	− 432.84	− 429.19	− 411.72	− 418.52	− 300.19
	Std	32.96	33.69	52.49	55.61	50.96
	<i>p</i> -value	0.4765	0.4689	0.8863	0.6987	0.4632
DE	Ave	− 431.89	− 425.92	− 410.34	− 415.29	− 295.64
	Std	40.29	41.96	55.61	58.85	60.29
	<i>p</i> -value	0.6196	0.6371	0.9126	0.7456	0.7698
Basic BBO-linear	Ave	− 430.51	− 420.46	− 408.83	− 411.96	− 293.89
	Std	43.69	48.36	62.81	60.36	63.85
	<i>p</i> -value	0.7216	0.8219	0.9348	0.8624	0.9168

The proposed TDMBBO algorithm is also compared with eight algorithms from CEC'2013, CEC'2014, and CEC'2015, and two more as reference algorithms. To access these algorithms, please see Molina et al. [49]: IPOP-CMA-ES, GAMPC, ICMAES-ILS, NBIPOP-CMA, DRMA, L-SHADE, GAAPPADE, MVMO14, SPS-L-SHADE-EIG, and L-SHADE-ND. For each test function and dimension, the algorithms are sorted by its average fitness function. Then, the average ranking for the different functions is calculated. For a better comparison, we show the rank of algorithms. Two nonparametric tests called Friedman and Hochberg have been also used to evaluate the performance of algorithms.

Tables 12, 13, 14 indicate the results of the algorithms in the CEC2005 problems. The algorithms are run 25 times for each problems. Tables 12, 13, 14 report the standard deviation of the fitness function (Std), the nonparametric

Table 13 The results of CEC2005 multimodal test functions

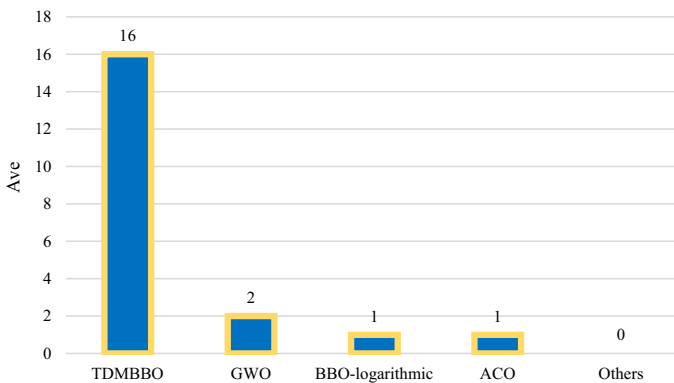
Algorithm	Metric	CEC _{F6}	CEC _{F7}	CEC _{F8}	CEC _{F9}	CEC _{F10}	CEC _{F11}	CEC _{F12}	CEC _{F13}	CEC _{F14}
TDMBBO	Ave	399.026	-175.96	-127.87	-316.72	-315.28	91.89	-459.33	-128.98	-279.56
	Std	7.26	0.76	8.72	9.62	5.19	0.21	0.0084	0.58	0.19
	<i>p</i> -value	N/A	N/A	0.0045	N/A	N/A	N/A	N/A	0.0026	N/A
GWO	Ave	401.19	-171.16	-128.26	-314.29	-313.73	92.93	-459.06	-127.24	-278.63
	Std	19.32	26.29	5.85	18.32	19.46	5.81	0.09	4.12	4.13
	<i>p</i> -value	0.0086	0.1863	N/A	0.0086	0.0037	0.0039	0.0011	0.0126	0.0089
BBO-logarithmic	Ave	400.89	-172.29	-125.33	-313.86	-310.19	93.95	-459.12	-129.08	-277.12
	Std	13.08	21.12	19.32	26.73	26.24	7.15	0.07	0.14	9.32
	<i>p</i> -value	0.0051	0.0896	0.0793	0.0374	0.1129	0.0198	0.0011	N/A	0.1236
ACO	Ave	403.32	-173.19	-123.09	-312.83	-312.96	94.76	-458.29	-125.29	-276.14
	Std	29.29	14.39	34.43	28.32	22.49	10.23	1.09	10.93	15.23
	<i>p</i> -value	0.1236	0.0086	0.2196	0.1423	0.0896	0.1025	0.0092	0.0852	0.1563
BBO-generalized sinusoidal	Ave	406.62	-168.76	-122.39	-310.73	-303.48	95.23	-457.38	-124.12	-274.09
	Std	35.32	39.26	39.29	35.49	62.12	18.18	5.77	14.01	22.34
	<i>p</i> -value	0.2543	0.4419	0.2286	0.2136	0.6685	0.2153	0.1089	0.1039	0.2463
BBO-sinusoidal	Ave	409.37	-167.38	-120.27	-308.76	-301.96	96.45	-455.38	-122.83	-273.26
	Std	44.32	51.09	44.26	44.18	68.85	25.83	10.38	20.08	29.24
	<i>p</i> -value	0.3312	0.4963	0.3126	0.4416	0.7136	0.2635	0.2019	0.2698	0.2863
BBO-quadratic	Ave	408.27	-165.29	-121.34	-306.37	-304.83	96.92	-452.96	-121.76	-271.39
	Std	41.36	55.21	41.83	50.29	48.25	26.86	25.36	30.24	36.75
	<i>p</i> -value	0.3156	0.5126	0.3076	0.6235	0.4029	0.2635	0.4269	0.3654	0.4962
DE	Ave	411.39	-165.09	-118.96	-307.96	-305.49	98.21	-453.74	-120.43	-275.96
	Std	50.35	53.46	48.52	48.37	41.28	28.63	21.38	33.26	18.36
	<i>p</i> -value	0.5396	0.5126	0.4236	0.5569	0.3684	0.3325	0.3576	0.4128	0.1932

Table 13 (continued)

Algorithm	Metric	CEC _{F6}	CEC _{F7}	CEC _{F8}	CEC _{F9}	CEC _{F10}	CEC _{F11}	CEC _{F12}	CEC _{F13}	CEC _{F14}
Basic BBO-linear	Ave	413.96	-163.73	-119.76	-305.86	-307.29	99.27	-451.86	-120.86	-270.42
	Std	52.39	59.45	45.84	55.91	36.85	30.16	29.16	32.26	45.13
	<i>p</i> -value	0.6953	0.6218	0.4028	0.7451	0.2698	0.4125	0.5129	0.4128	0.5237

Table 14 The results of CEC2005 composite test functions

Algorithm	Metric	CEC _{F15}	CEC _{F16}	CEC _{F17}	CEC _{F18}	CEC _{F19}	CEC _{F20}
TDMBBO	Ave	175.46	169.32	211.96	50.96	45.91	64.14
	Std	33.12	37.16	48.36	21.08	15.89	21.09
	<i>p</i> -value	N/A	N/A	0.0091	N/A	N/A	0.0049
GWO	Ave	180.19	173.14	210.36	56.13	48.81	67.14
	Std	48.36	41.08	45.91	33.19	24.08	30.17
	<i>p</i> -value	0.0886	0.0256	N/A	0.0091	0.0081	0.0175
BBO-logarithmic	Ave	190.36	172.19	218.10	58.14	46.86	68.76
	Std	55.36	38.16	60.15	35.16	20.19	37.17
	<i>p</i> -value	0.2863	0.0086	0.1036	0.0756	0.0038	0.0283
ACO	Ave	195.38	180.12	220.14	60.13	51.14	62.76
	Std	60.25	54.16	63.38	40.72	26.18	19.23
	<i>p</i> -value	0.3239	0.1136	0.1963	0.1029	0.0453	N/A
BBO-generalized sinusoidal	Ave	200.29	183.19	222.93	65.19	57.18	75.14
	Std	65.14	60.65	68.19	51.79	41.20	43.28
	<i>p</i> -value	0.5123	0.2236	0.2456	0.4123	0.3285	0.1756
BBO-sinusoidal	Ave	204.43	185.09	225.91	73.12	65.19	80.18
	Std	71.36	65.43	74.32	59.17	51.83	47.13
	<i>p</i> -value	0.5532	0.3109	0.3329	0.5236	0.4156	0.3126
BBO-quadratic	Ave	208.96	184.39	228.19	76.91	62.11	85.17
	Std	74.63	63.14	79.21	64.29	61.29	53.72
	<i>p</i> -value	0.6238	0.3029	0.4562	0.6129	0.3862	0.4236
DE	Ave	212.86	188.96	227.19	80.29	72.96	88.91
	Std	82.63	70.29	76.19	71.29	75.14	59.2
	<i>p</i> -value	0.7126	0.4569	0.4321	0.6895	0.5169	0.4786
Basic BBO-linear	Ave	218.86	193.24	230.59	80.91	77.18	92.34
	Std	83.21	76.15	82.17	85.16	86.17	68.19
	<i>p</i> -value	0.8426	0.6629	0.5523	0.7689	0.7129	0.5326

**Fig. 16** The ranking of algorithms at “Ave” metric on CEC2005 problems

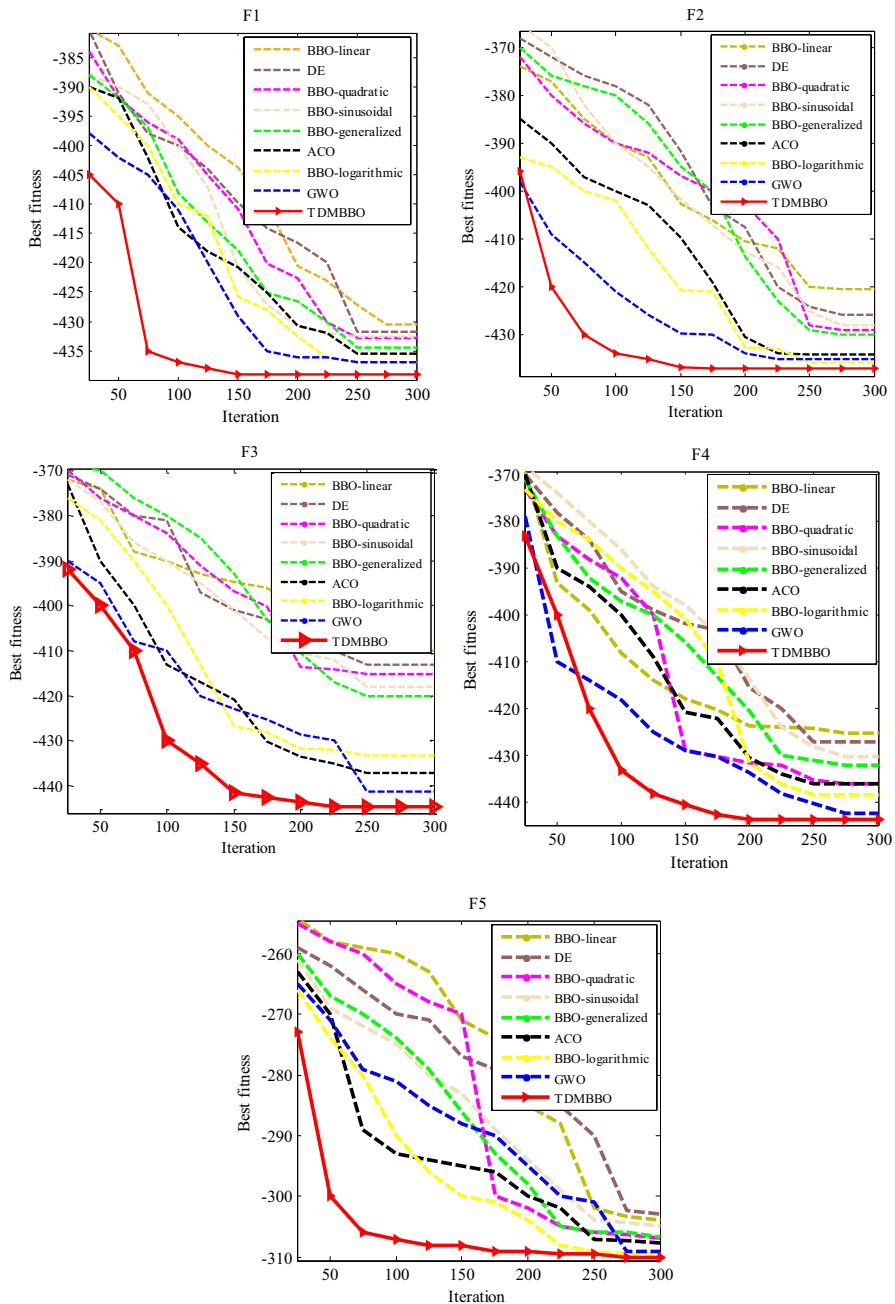


Fig. 17 The convergence curve of algorithms on unimodal CEC2005 test functions

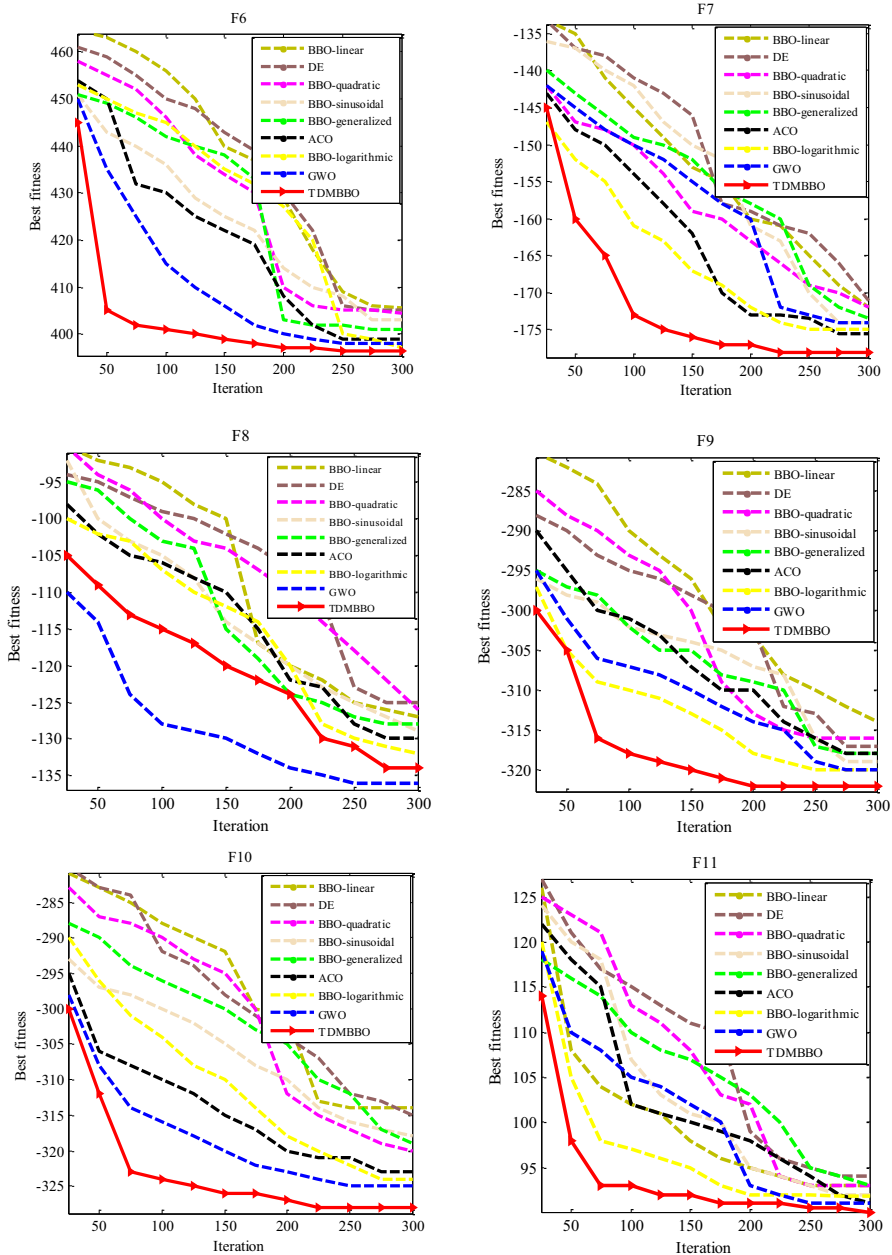


Fig. 18 The convergence curve of algorithms on multimodal CEC2005 test functions

Wilcoxon's test (p -value), and the mean value of the fitness function (Ave). It should be noted that the best outcomes are reported in bold kind. Wilcoxon's test was performed at a significance level of 5.00%. In this test, N/A stands for

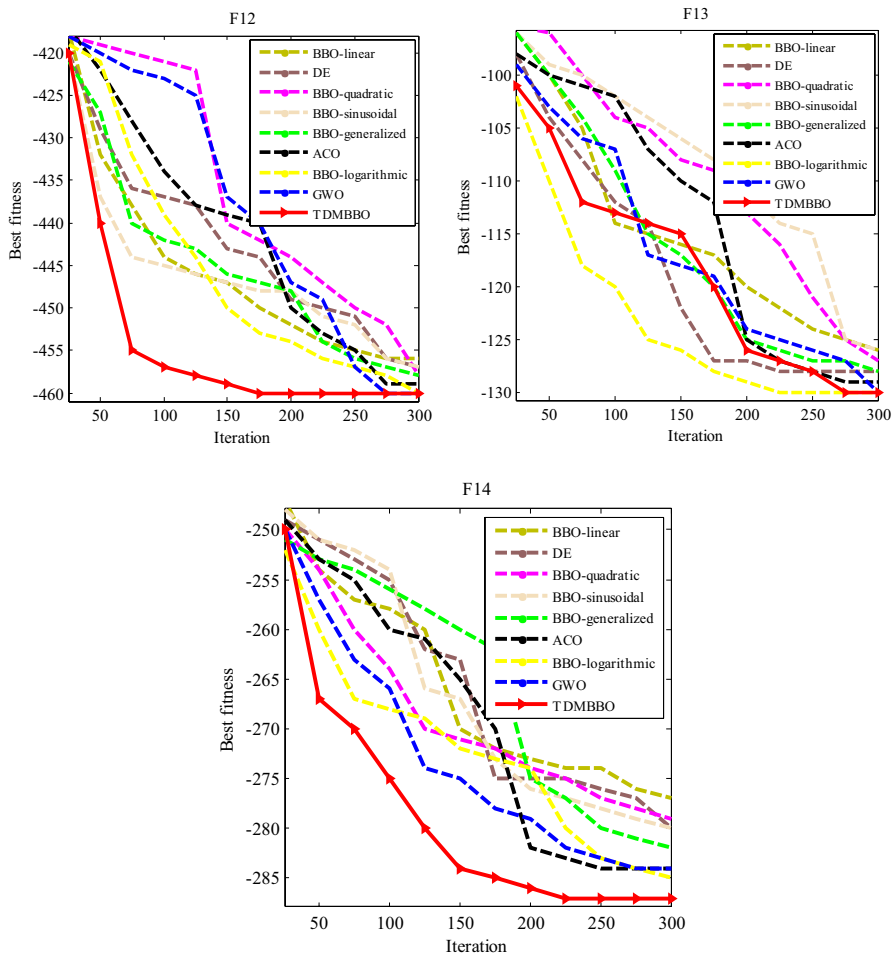


Fig. 18 (continued)

"Not Applicable," meaning that the TDMBBO cannot be statistically compared to itself. As can be seen, the results of the TDMBBO are better than the other algorithms.

Table 12 shows the results of unimodal CEC2005 test functions. In this table, TDMBBO has reached the highest "Ave" value in 5 functions (100.00%). Table 13 shows the results of CEC2005 multimodal test functions. In this table, TDMBBO has reached the highest "Ave" value in 7 functions (77.77%). Table 14 also shows the results of CEC2005 composite test functions. In this table, TDMBBO has reached the highest "Ave" value in 4 functions (66.66%). Given the standard deviation and p -values of algorithms, TDMBBO can achieve significant results compared to other algorithms. Figure 16 indicates the performance of the algorithms at the "Ave" metric. Figures 17, 18, 19 show the convergence curve of

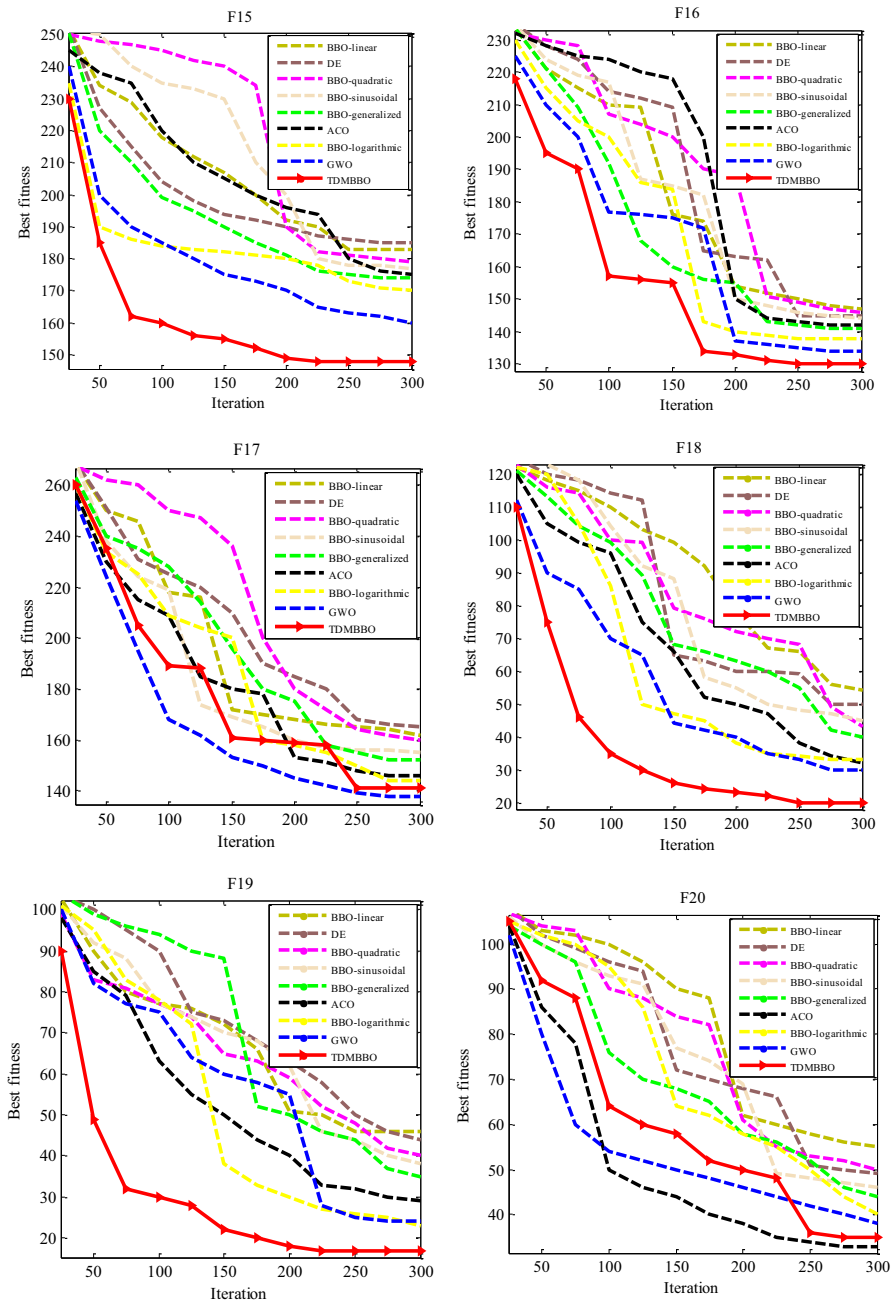


Fig. 19 The convergence curve of algorithms on composite CEC2005 test functions

Table 15 The rank of algorithms for IEEE CEC test functions

Algorithm	CEC'2013			CEC'2014			CEC'2015			Average rank
	D10	D30	D50	D10	D30	D50	D10	D30	D50	
IPOP-CMAES	8	8	9	8	6	6	10	8	9	7.89
GAMPC	11	11	11	11	11	11	11	11	11	11
ICMAES-ILS	5	6	5	5	4	1	8	7	7	5.34
NBIPOPaCMA	10	10	10	10	9	9	9	9	8	9.34
DRMA	6	7	7	6	7	8	5	6	4	5.23
L-SHADE	2	2	3	3	2	2	6	5	6	3.45
GAAPPADE	4	4	6	9	8	7	1	3	2	4.89
MVMO14	9	9	8	7	10	10	7	10	10	8.89
SPS-L-SHADE-EIG	3	5	4	4	5	3	3	1	3	3.45
L-SHADE-ND	7	1	2	2	3	5	4	4	5	3.67
TDMBBO	1	3	1	1	1	4	2	2	1	1.78

Table 16 The results of algorithms in Hochberg test (vs. TDMBBO)

Algorithm	CEC'2013 (D10)		CEC'2014 (D30)		CEC'2015 (D50)	
	Unadjusted P	PHochberg	Unadjusted P	PHochberg	Unadjusted P	PHochberg
IPOP-CMAES	0.098523	0.798521	0.010258	0.096321	0.000845	0.006452
GAMPC	0.000002	0.000019	0.000002	0.000018	0.000000	0.000001
ICMAES-ILS	0.582145	0.803654	0.091542	0.156981	0.008526	0.092145
NBIPOPaCMA	0.003186	0.027215	0.000298	0.001596	0.001789	0.008452
DRMA	0.283248	0.801083	0.023651	0.096321	0.095212	0.102365
L-SHADE	0.806215	0.803654	0.452147	0.524265	0.009521	0.092145
GAAPPADE	0.615465	0.803654	0.009752	0.014231	0.423651	0.496521
MVMO14	0.024985	0.178693	0.000026	0.000085	0.000005	0.000009
SPS-L-SHADE-EIG	0.765852	0.803654	0.000941	0.006254	0.365218	0.496521
L-SHADE-ND	0.202586	0.801083	0.236512	0.256981	0.009851	0.092145

algorithms on CEC2005 functions. The convergence curve of the TDMBBO is faster than other algorithms. TDMBBO achieved the best convergence curve in 16 test functions.

Table 15 shows the ranking of algorithms for CEC'2013–2015 benchmark problems with different dimensions. The best ranking of algorithms is marked in bold. In the CEC'2013 benchmark, TDMBBO, L-SHADE-ND, and TDMBBO have the best rank in D10, D30, and D50, respectively. In the CEC'2014 benchmark, TDMBBO, TDMBBO, and ICMAES-ILS have the best rank in D10, D30, and D50, respectively. In the CEC'2015 benchmark, GAAPPADE, SPS-L-SHADE-EIG, and TDMBBO have the best rank in D10, D30, and D50, respectively. The last column of

Table 17 The details of IEEE CEC06-2019 test functions

No	Functions	$F_t^* = F_t(x^*)$	D	Search range
F1	Storn's chebyshev polynomial fitting problem	1	9	[− 8192, 8192]
F2	Inverse hilbert matrix problem	1	16	[− 16384, 16384]
F3	Lennard–Jones minimum energy cluster	1	18	[− 4, 4]
F4	Rastrigin's function	1	10	[− 100, 100]
F5	Griewank's function	1	10	[− 100, 100]
F6	Weierstrass function	1	10	[− 100, 100]
F7	Modified Schwefel's Function	1	10	[− 100, 100]
F8	Expanded Schaffer's F6 Function	1	10	[− 100, 100]
F9	Happy cat function	1	10	[− 100, 100]
F10	Ackley's function	1	10	[− 100,100]

Table 15 also shows the average rankings of the algorithms across all problems. As can be seen, TDMBBO produced the most favorable results (average rank = 1.78). SPS-L-SHADE-EIG, L-SHADE, L-SHADE-ND, and GAAPPADE are also in the next ranks, respectively. GAMPC has the worst results.

Table 16 shows the statistical results of the algorithms in the Hochberg test. The first column is the original p -value, and the second column is the adjusted p -value obtained by the Hochberg method. In the CEC'2013, GAMPC and L-NBIPOPACMA had the worst results in D10. In the CEC'2014, GAMPC,

Table 18 The results of algorithms on IEEE CEC06-2019 test functions

Rank	Algorithm	Test functions										Total
		F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	
1	TDMBBO	10	10	10	10	10	10	10	10	9.33	10	99.33
2	I-CSA	10	10	10	10	10	10	10	8.15	8.26	10	96.41
3	CapSA	10	10	10	10	10	10	10	7.36	8.17	10	95.53
4	IG	10	10	10	10	10	10	9.85	6.39	5.69	10	91.80
5	GVNS	10	10	10	10	10	10	8.59	4.96	5.08	10	88.63
6	ChOA	10	10	10	10	10	10	7.33	5.29	4.86	10	87.48
7	GWO	10	10	10	10	10	10	3.69	5.71	3.85	10	83.25
8	BBO-logarithmic	10	10	10	10	10	9.56	4.29	3.29	2.95	10	80.09
9	BBO-generalized sin	10	10	10	10	10	8.96	4.45	3.09	2.51	10	79.01
10	BBO-sinusoidal	10	10	10	10	10	7.62	3.98	2.50	2.08	10	76.18
11	BBO-quadratic	10	10	10	10	8.26	10	2.09	2.26	2.12	10	74.73
12	BBO-linear	10	10	10	10	10	5.98	3.26	1.96	1.36	10	72.56
13	PSO	10	10	10	10	10	5.69	2.78	1.85	1.12	10	71.44
14	GA	10	10	10	6.96	10	10	1.96	1.09	1.28	10	71.29

GAAPPADE, MVMO14, SPS-L-SHADE-EIG, and L-NBIPOPcMA had the worst results in D30. In the CEC'2015, GAMPC, IPOP-CMAES, NBIPOPcMA, and MVMO14 have the worst results in D50.

3.5 IEEE CEC06-2019

In this part, to evaluate the efficiency of the proposed TDMBBO, IEEE CEC06-2019 test functions are used. Table 17 shows the details of IEEE CEC06-2019. The more details of IEEE CEC06-2019 test functions are available in Kaidi et al. [17]. The 3D visualization of these benchmark functions is showed in Fig. 25.

Table 18 shows the results of algorithms on the IEEE CEC06-2019 test functions. TDMBBO shows the best result in 9 out of 10 functions. I-CSA and CapSA algorithms represent the best results in 8 and 8 out of 10 problems, respectively. GA and PSO got the worst results. In most previous migration models, one function was used for the migration rate. The main disadvantage of these models is that all habitats are treated the same way, and emigration and immigration rates follow a single mathematical function. In TDMBBO, for the migration rates of each habitat with low, medium, and high HSI, a special mathematical model is considered that can apply the appropriate migration rate.

3.6 Evaluation of TDMBBO with previous studies

In this part, the performance of the TDMBBO has been compared with that of six other algorithms: weighted superposition attraction (WAS) [50], the best performance algorithm (BPA), the iterative best performance algorithm (IBPA), the largest absolute difference algorithm (LADA), the tabu search (TS), and simulated annealing (SA) [51]. The result of these six algorithms is categorized into Baykasoğlu and Akpinar [50]. Here, the TDMBBO has been implemented 30 times for each function. The statistical results of the algorithms have been displayed in Table 19. The 3D visualization of 30 functions is presented in Fig. 26.

According to Table 19, TDMBBO is successful in bringing out the value of “Best” in about 21 out of 33 functions (63.63%). WSA in 17 functions (51.51%), BPA in 6 functions (18.18%), IBPA in 5 functions (15.15%), LADA in 4 functions (12.12%), and TS and SA in 3 functions (9.09%) are successful in bringing out the value of “Bbest.” In four problems (EM, RG, RB, and SWF), TDMBBO could not produce good results, which are 12/12% of the benchmark set. In addition, the standard deviation values (Std-Dev) in Table 19 show that TDMBBO is more stable than other algorithms.

3.7 Real-world engineering optimization problems

In this section, two real-world engineering optimization problems are used to evaluate the algorithms. Due to the complexity of these problems, solving them can be a challenge for meta-heuristic algorithms.

Table 19 Computational results for the test functions

Function	$F(x^*)$	Results	BPA	IBPA	LADA	TS	SA	WSA	TDMBBO
F1 (ACK)	0	Best	4.11E-3	8.15E-3	8.82E-3	1.4185E-1	1.261E-2	0.000	1.074E-10
		StdDev	1.338E-2	1.021E-2	1.57E-3	7.488E-3	5.456E-2	0.000	0.00
F2 (AP)	= -0.35239	Best	-3.5238E-1	-3.5238E-1	-3.5238E-1	-3.5238E-1	-3.5238E-1	-3.5238E-1	-0.35239
		StdDev	1.4490E-6	1.0670E-6	5.5760E-7	2.1830E-5	6.6540E-5	8.7610E-6	0.00
F3 (BL)	0	Best	1.0170E-8	3.2170E-9	1.2590E-9	3.9550E-7	1.3740E-6	5.5890E-8	0.00
		StdDev	2.3230E-7	2.8380E-7	2.7040E-7	6.3020E-6	2.9700E-5	3.8770E-8	0.00
F4 (BF1)	0	Best	3.6560E-5	2.3620E-7	6.6280E-6	1.1230E-4	2.5920E-5	0.000	0.00
		StdDev	1.59E-3	7.6110E-4	1.22E-3	1.801E-2	3.06E-3	0.000	0.00
F5 (BF2)	0	Best	2.9270E-6	3.3970E-6	4.9110E-5	6.2230E-4	1.5890E-4	0.000	0.00
		StdDev	1.44E-3	3.976E-2	6.43E-3	2.656E-2	6.67E-3	0.000	0.00
F6 (BP)	(=0.39789)	Best	3.9788E-1	3.9788E-1	3.9788E-1	3.9788E-1	3.9788E-1	3.9788E-1	0.39789
		StdDev	2.3610E-6	2.1560E-6	9.3730E-5	3.5430E-5	8.4430E-5	9.4750E-6	0.00
F7 (CB3)	0	Best	6.0350E-10	5.4050E-9	7.5220E-9	1.3850E-8	1.1740E-6	0.000	0.00
		StdDev	6.2550E-7	3.1230E-7	1.7520E-6	1.4370E-5	3.4580E-5	0.000	0.00
F8 (CB6)	= -1.03163	Best	-0.103162E1	-0.103162E1	-0.103162E1	-0.103162E1	-0.103162E1	-0.103161E1	-1.03163
		StdDev	2.2550E-6	2.2210E-6	1.7470E-6	2.0600E-5	5.6150E-5	4.5160E-16	0.00
F9 (CM)	0.2	Best	1.9999E-1	1.9999E-1	1.9999E-1	1.9999E-1	1.9999E-1	0.200	0.2
		StdDev	2.6470E-7	1.6460E-7	6.4310E-8	4.2710E-6	3.3090E-5	8.4690E-17	0.00
	0.4	Best	3.9999E-1	3.9999E-1	3.9987E-1	3.9994E-1	3.9991E-1	0.400	0.4
		StdDev	4.3540E-6	4.9510E-6	6.9090E-4	3.6040E-4	7.0500E-4	1.6930E-16	0.00
F10 (DA)	-24.777	Best	-24.776.510	-24.776.510	-24.776.510	-24.776.510	-24.776.510	-24.776.520	-24.776.51
		StdDev	5.370E-2	5.579E-2	2.5194E-1	2.7919E-1	9.251E-2	1.570E-2	0.04231
F11 (EP)	-1	Best	-9.9999E-1	-9.9999E-1	-9.9999E-1	-9.9999E-1	-9.9999E-1	-9.9999E-1	-1
		StdDev	2.0030E-6	3.79001E-1	2.8850E-6	5.0733E-1	1.0110E-4	2.0250E-4	4.945E-3
F12 (EM)	-9.660152	Best	-0.955986E1	-0.952608E1	-0.927356E1	-0.897041E1	-0.952701E1	-0.720847E1	-7.45631
		StdDev	2.1804E-1	1.6721E-1	2.6812E-1	1.3066E-1	2.2567E-1	7.655710	0.26471

Table 19 (continued)

Function	$F (x^*)$	Results	BPA	IBPA	LADA	TS	SA	WSA	TDMBBO
F13 (EXP)	1	Best	9.9998E-1	9.9998E-1	9.9986E-1	9.9994E-1	9.9207E-1	1.000	1.00
		StdDev	2.4690E-5	1.6340E-5	1.1230E-4	7.9750E-4	4.46E-3	0.000	0.00
F14 (GP)	3	Best	3.00000	3.00000	3.00000	3.00000	3.00000	3.00000	3.00000
		StdDev	2.5860E-5	8.238470	16.466700	5.7510E-4	1.140E-3	1.6220E-4	0.00
F15 (GRP)	0	Best	1.2080E-6	5.3990E-6	8.1240E-5	3.1200E-5	2.3520E-5	32.830	1.173E-4
		StdDev	0.001300	0.001620	3.4560E-4	1.3820E-4	2.2220E-4	1.4450E-15	4.241E-3
F16 (HSK)	-2.3458	Best	-2.345810	-2.345810	-2.345810	-2.345810	-2.345810	-2.345810	-2.34581
		StdDev	8.2030E-8	1.3560E-7	5.3270E-8	3.1690E-6	1.9940E-5	1.6310E-4	2.43E-8
F17 (KL)	3.0748E-4	Best	3.0750E-4	3.0750E-4	3.1050E-4	3.0750E-4	3.0870E-4	3.0790E-4	3.075E-4
		StdDev	3.4880E-6	4.0960E-6	6.4000E-5	6.6380E-7	6.8420E-6	1.7060E-6	4.341E-10
F18 (LM1)	0	Best	1.0050E-7	1.0950E-7	8.2400E-8	3.1240E-6	3.5150E-5	3.3150E-7	3.362E-12
		StdDev	6.0410E-6	6.7160E-6	2.8670E-6	1.1860E-4	2.2570E-4	3.3920E-6	1.045E-10
F19 (LM2)	0	Best	1.38E-3	1.882E-2	1.8022E-1	1.7907E-1	7.06E-3	0.0103E-3	1.981E-07
		StdDev	2.7903E-1	4.8275E-1	6.7992E-1	3.2601E-1	2.5005E-1	1.237E-2	2.513E-11
	0	Best	3.87E-3	9.62E-3	6.966E-2	5.080E-2	1.122E-2	2.570E-3	0.00334
		StdDev	4.0486E-1	4.1149E-1	4.9381E-1	2.4229E-1	2.8039E-1	5.220E-2	0.01231
F20 (MR)	4.358E-5	Best	4.3560E-5	4.3580E-5	4.4260E-5	4.3570E-5	4.5740E-5	4.4090E-5	1.741E-02
		StdDev	3.0540E-5	2.2700E-5	2.2640E-5	7.2990E-6	1.0270E-5	2.2710E-6	1.341E-08
F21 (MCP)	0	Best	6.6840E-12	5.6060E-12	5.2600E-8	3.3760E-11	1.0980E-9	8.9340E-13	0.00
		StdDev	8.5760E-9	1.2350E-8	3.1290E-6	2.1670E-9	5.1130E-7	1.7530E-10	0.00
F22 (MRP)	0	Best	1.9600E-6	5.8650E-7	3.2100E-8	1.6230E-5	1.7320E-5	4.2910E-9	0.00
		StdDev	2.37E-3	4.030E-3	2.180E-3	7.948E-4	1.0370E-4	3.8710E-7	3.874E-4
F23 (MGP)	1.29695	Best	1.296950	1.296950	1.296950	1.296940	1.296940	1.296950	1.29695
		StdDev	4.8880E-6	3.605E-2	2.033E-2	1.2860E-4	3.4700E-4	1.7780E-4	1.234E-3

Table 19 (continued)

Function	$F(x^*)$	Results	BPA	IBPA	LADA	TS	SA	WSA	TDMBBO
F24 (PWQ)	0	Best	1.6510E-5	2.2940E-7	3.5300E-4	1.320E-3	1.1130E-4	0.000	0.00
		StdDev	1.330E-3	1.360E-3	3.693E-2	8.970E-3	3.300E-3	0.000	0.00
F25 (RG)	0	Best	1.6961E-1	8.790E-2	6.060E-3	4.587530	3.932E-2	0.000	4.2145
		StdDev	1.6469E-1	1.2481E-1	5.540E-3	8.9403E-1	9.522E-2	0.000	2.14532
F26 (RB)	0	Best	3.300500	1.657800	13.11610	24.73950	9.1870E-1	8.91670	4.1475
		StdDev	17.97070	14.92020	14.95210	19.17630	1.81800	1.600E-2	2.4123
F27 (SAL)	0	Best	9.987E-2	9.987E-2	9.987E-2	9.996E-2	9.987E-2	0.000	0.09987
		StdDev	9.609E-2	7.301E-2	4.966E-2	8.101E-2	9.040E-2	0.000	0.00145
	0	Best	2.9988E-1	2.9987E-1	3.0989E-1	3.152E-2	3.0006E-1	0.000	0.21423
		StdDev	8.947E-2	1.2510E-1	8.872E-2	1.6112E-1	1.5625E-1	0.000	0.12041
F28 (SF1)	0	Best	1.20510E-4	1.9790E-5	1.5350E-6	1.4090E-4	6.6710E-4	0.000	0.00
		StdDev	3.620E-3	4.740E-3	4.440E-3	3.430E-3	2.770E-3	0.000	0.00
F29 (SF2)	0	Best	6.332E-2	8.329E-2	2.1116E-1	5.8920E-1	6.096E-2	8.93710E-7	1.341E-05
		Mean	1.6367E-1	3.217740	3.248470	3.633560	3.0990E-1	1.20110E-6	1.341E-05
F30 (SWF)	-4189.829	Best	-4188.830	-4188.800	-4189.790	-4146.860	-4189.640	-4080.130	-3845.3245
		StdDev	1.417130	9.0048E-1	3.6700E-2	23.245700	2.3796E-1	328.200	7214.2417

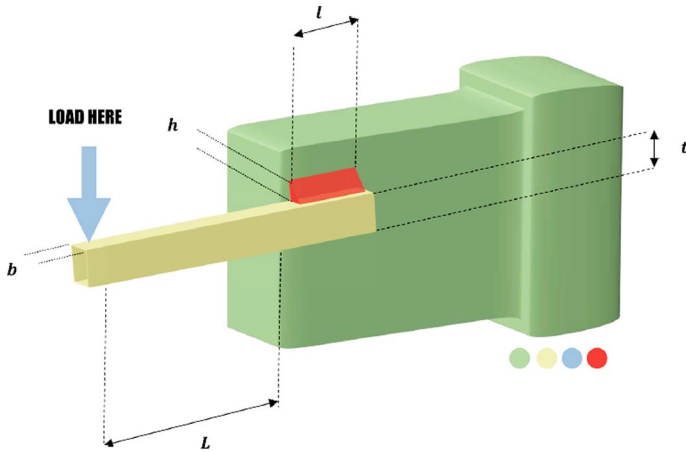


Fig. 20 Welded beam system

3.7.1 Welded beam system

In this section, the well-known welded-beam problem is used to evaluate TDMBBO. Here, the goal is to minimize the total cost of construction. A schematic view of the problem is shown in Fig. 20 and can be formulated as Eq. (14)–(22) [29].

$$f(X) = 1.10471x_1^2x_2 + 0.04811x_3x_4(14 + x_2) \quad (14)$$

Subjected to

$$g_1(X) = \tau(x) - \tau_{max} \leq 0 \quad (15)$$

$$g_2(X) = \sigma(x) - \sigma_{max} \leq 0 \quad (16)$$

$$g_3(X) = x_1 - x_4 \leq 0 \quad (17)$$

$$g_4(X) = 0.10471x_1^2 + 0.04811x_3x_4(14 + x_2) - 5 \leq 0 \quad (18)$$

$$g_5(X) = 0.125 - x_1 \leq 0 \quad (19)$$

$$g_6(X) = \delta(x) - \delta_{max} \leq 0 \quad (20)$$

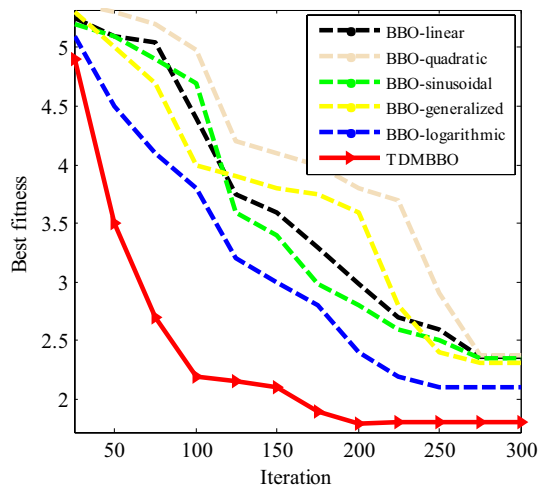
Table 20 Comparison of the results of algorithms for the welded beam problem

Algorithm	Best fitness	Mean fitness	Standard deviation	Run time (s)
TDMBBO	1.80152	1.80308	0.0000041	206
I-CSA	2.04239	2.51269	0.0007563	239
CapSA	2.02519	2.41265	0.0006325	225
IG	2.16321	2.85362	0.0025631	221
GVNS	2.23654	2.93214	0.0325869	269
ChOA	2.09314	2.62143	0.0016521	251
GWO	2.20317	2.85263	0.0187453	249
ACO	2.28965	3.01562	0.1578923	296
DE	2.33546	3.85624	0.1856231	281
BBO-logarithmic	2.10562	2.90562	0.0426893	241
BBO-generalized sin	2.31652	3.45695	0.1698524	225
BBO-sinusoidal	2.35635	3.92654	0.3542198	231
BBO-quadratic	2.37895	3.99652	0.4965231	229
BBO-linear	2.36527	3.77856	0.3356954	215
PSO	2.37896	4.12569	0.9856254	309
GA	2.39856	4.26531	1.5589621	268

$$g_7(X) = P - P_c(x) \leq 0 \quad (21)$$

$$0.1 \leq x_1, x_4 \leq 2, 0.1 \leq x_2, x_3 \leq 10 \quad (22)$$

Where

Fig. 21 The convergence curve of algorithms for welded beam problem

$$\begin{aligned}
\tau(x) &= \sqrt{(\tau')^2 + 2\tau'\tau''\frac{x_2}{2R} + (\tau'')^2} \\
\tau' &= \frac{P}{\sqrt{2x_1x_2}} \\
\tau'' &= \frac{PMR}{J} \\
M &= P\left(L + \frac{x_2}{2}\right) \\
J &= 2\left\{\sqrt{2x_1x_2}\left[\frac{x_2^2}{12} + \left(\frac{x_1+x_3}{2}\right)^2\right]\right\} \\
x_1 &= \text{The thickness of the weld} \\
x_2 &= \text{The length of the welded joint}
\end{aligned}
\quad
\begin{aligned}
\sigma(x) &= \frac{6PL}{x_3^2x_4} \\
\delta(x) &= \frac{4PL^3}{Ex_3^2x_4} \\
P_c(x) &= \frac{4.013\sqrt{E(x_3^2x_4/36)}}{L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right) \\
R &= \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1+x_3}{2}\right)^2} \\
P &= 6000lb, L = 14in, E = 30 \times 10^6psi, G = 12 \times 10^6psi, \\
\tau_{max} &= 13600psi, \sigma_{max} = 30000psi, \text{ and } \delta_{max} = 0.25in. \\
x_3 &= \text{The width of the beam} \\
x_4 &= \text{The thickness of the beam}
\end{aligned}$$

Table 20 shows the results of algorithms on the welded beam problem. As can be seen, TDMBBO has obtained the best value of the fitness function compared to other algorithms. The optimal fitness function of TDMBBO is 1.8014. The standard deviation values in Table 20 show that TDMBBO is more stable than other algorithms. Figure 21 shows the convergence curve of TDMBBO and other BBO algorithms. According to Fig. 21, the convergence curve of the TDMBBO is faster than others. According to Table 20, the runtime of TDMBBO is less than other algorithms. The runtime of all models of the BBO is appropriate.

3.7.2 Pressure vessel problem

Optimizing the cost of producing a pressure vessel with two parabolic heads is chosen for evaluating the TDMBBO's performance. The schematic view of the problem is shown in Fig. 22 and can be formulated as Eq. (23)–(28) [31].

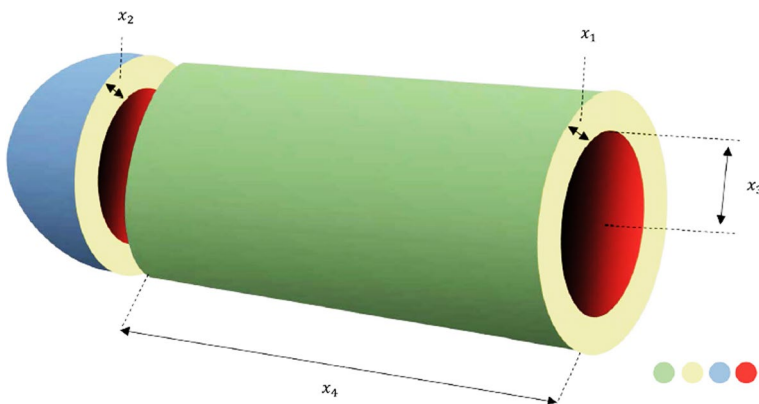
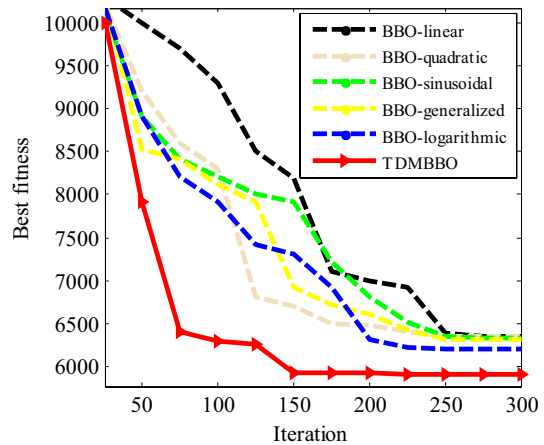


Fig. 22 The pressure vessel system

Table 21 The results of the algorithms for pressure vessel system

Algorithm	Best fitness (\$)	Mean fitness (\$)	Standard deviation	Run time (s)
TDMBBO	5915.2463	5921.2469	0.0008145	226
I-CSA	6055.1296	6174.2365	0.0198563	240
CapSA	6079.2541	6274.8961	0.2196524	269
IG	6124.2898	6359.7521	1.9856321	238
GVNS	6112.8561	6317.4698	1.3856323	291
ChOA	6175.7469	6459.1796	3.8635215	283
GWO	6219.4963	6529.9527	6.4631587	274
ACO	6208.2983	6489.2874	5.2456321	328
DE	6288.9654	6695.8256	8.3952614	302
BBO-logarithmic	6196.2563	6523.6981	2.9563214	251
BBO-generalized sin	6309.3652	6985.4632	15.369854	247
BBO-sinusoidal	6325.2451	7029.2541	18.965214	249
BBO-quadratic	6340.5963	7196.5259	21.965237	236
BBO-linear	6352.9652	7265.9567	25.365478	232
PSO	6218.9586	6659.8419	6.0896523	309
GA	6339.9652	7096.1478	20.856952	293

Fig. 23 The convergence trend of algorithms for pressure vessel system

$$f(X) = 0.6224x_1x_3x_4 + 1.7781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3 \quad (23)$$

Subjected to

$$g_1(X) = -x_1 + 0.0193x_3 \leq 0 \quad (24)$$

$$g_2(X) = -x_2 + 0.00954x_3 \leq 0 \quad (25)$$

$$g_3(X) = -\pi x_3^2 x_4 - \frac{4}{3} \pi x_3^3 + 129600 \leq 0 \quad (26)$$

$$g_4(X) = x_4 - 240 \leq 0 \quad (27)$$

$$0 \leq x_1, x_2 \leq 100, \quad 10 \leq x_3, x_4 \leq 200 \quad (28).$$

Where x_1 , The shell thickness (T_s). x_2 , The head thickness (T_h). x_3 , The radius of the cylindrical shell (R). x_4 , The length of the shell (L).

Table 21 shows the results of the algorithms for the pressure vessel problem. The best value of the fitness function obtained by the TDMBBO is 5915/2463 \$, which is better than the results of other algorithms. Such a good result is achieved in only 300 iterations, which confirms the great convergence of the TDMBBO (Fig. 23). According to Table 21, the runtime of TDMBBO is less than other algorithms.

3.8 Discussion and the analysis of the results

In p-median, IEEE CEC benchmarks, and real-world problems, the TDMBBO converged earlier than other meta-heuristic algorithms (such as CapSA, ChOA, GWO, ACO, DE, PSO, GA, etc.). Since the migration in TDMBBO is such that a host habitat sends its habitants to a guest habitat, the fitness function of all habitats is improved. Therefore, the convergence trend is improved. Also, the TDMBBO converged earlier than the other BBO algorithms with previous migration models. In fact, in the proposed TDMBBO, for emigration and immigration rates, a particular mathematical model is utilized so that the algorithm can apply appropriate migration rates for different solutions. In TDMBBO, non-linear migration rates based on quadratic, cubic, sinusoidal, and hyperbolic tangent functions have been proposed. In most of the other BBO algorithms, one function for the migration rate has been used. The main disadvantage of these models is that emigration and immigration rates follow a single mathematical function.

One of the effective parameters of the proposed TDMBBO is the threshold p . The thresholds represent the points at which emigration and immigration rates change their migration functions. In other words, this parameter can be the optimal moment for changing the functions of the migration rates. This is another advantage of TDMBBO compared to other BBO algorithms.

In TDMBBO, when HSI is low, a quadratic function is used. This is because habitats with a lower HSI should have a much lower emigration rate and fewer species that are sent to other habitats. The result of using this logic is a more severe avoidance of weak migrations, which ultimately greatly improves the algorithm's convergence speed. The hyperbolic tangent has also a much higher emigration rate than other functions. In this function, the chances of selecting elite habitats to produce new habitats are equal. In an ascending hyperbolic tangent, the convergence rate of the algorithm will be improved. The immigration rate of the cubic function is higher than other functions and slowly decreases from its maximum value. The combination of an elite habitat and a very poor habitat always expands the neighborhood's search. The advantage of this model is to eliminate the worry of getting stuck in local minimums and increases the possibility of achieving the best solution. When HSI is high, the selection of an elite habitat as the host habitat through its immigration rate is not very useful and even reduces the convergence rate of algorithm. In TDMBBO, the immigration rate of the hyperbolic tangent function is lower than other functions and changes slowly. Therefore, due to these proposed functions, TDMBBO has higher exploitation and exploration power than other BBO algorithms.

According to the results, the runtime of TDMBBO is less than other algorithms. The runtime of all models of the BBO is appropriate. But TDMBBO has a shorter runtime to find the optimal solution compared to its previous models. [Table 9](#) shows the implementation of algorithms upon large random instances. In these instances, TDMBBO, IG, I-CSA, GVNS, CapCA, ChOA, GWO, PSO, and GA obtained the optimal solution in 8, 5, 5, 4, 6, 4, 4, 3, and 2 instances, respectively. TDMBBO has a high capability in all three types of random instances. As a result, TDMBBO still has powerful results when the complexity of the problem increases. Whereas other algorithms failed to determine the global optimal in all large random instances.

4 Conclusion and future works

Choosing a suitable location for emergency centers in developed cities plays an important role in maintaining people's tranquility. Therefore, it is necessary to select a number of optimal centers with the appropriate spatial distribution. The current study investigated the development of BBO with GIS capabilities to select appropriate sites for emergency centers in Tehran's district 2. A novel three-dimensional migration model of BBO (TDMBBO) has been introduced to optimize the p-median problem. In TDMBBO, nonlinear migration rates based on quadratic, cubic, sinusoidal, and hyperbolic tangent functions are proposed. The behavior of

TDMBBO has been examined on two location-allocation datasets, the IEEE CEC benchmark problems, three randomly generated datasets, and two real world optimization problems.

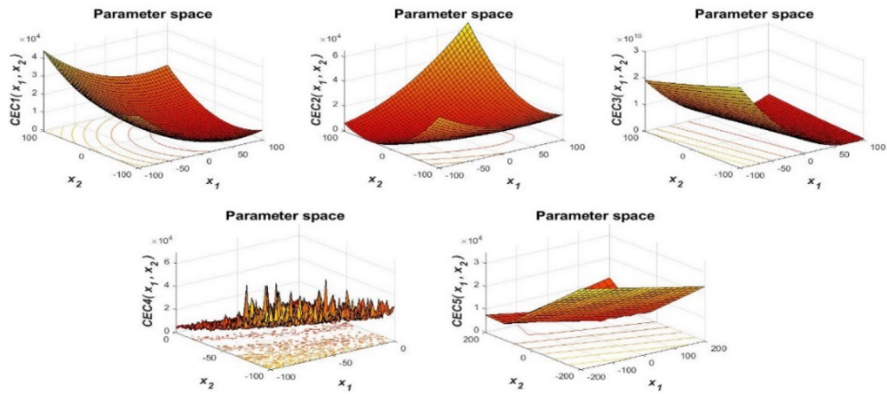
To evaluate the performance of TDMBBO, 31 competitive algorithms have been used. In general, TDMBBO has shown good performance in the location-allocation problem. The convergence process of the TDMBBO is much faster than that of the other algorithms. Also, TDMBBO has shown strong performance in benchmark problems. In IEEE CEC2005 problems, TDMBBO showed rapid convergence compared to other algorithms. The results show the TDMBBO's superiority and this algorithm's capability in solving real-world optimization problems.

In this paper, BBO has been developed using nonlinear models of migration rates. In general, developing an algorithm should be done based on nature-inspired mathematical logic before implementation. Here, six mathematical functions have been applied to model the migration operator of the BBO algorithm. Also, the experimental results of the proposed TDMBBO show that the algorithm can be very useful. Therefore, it can be said that the improvement of BBO is reliable.

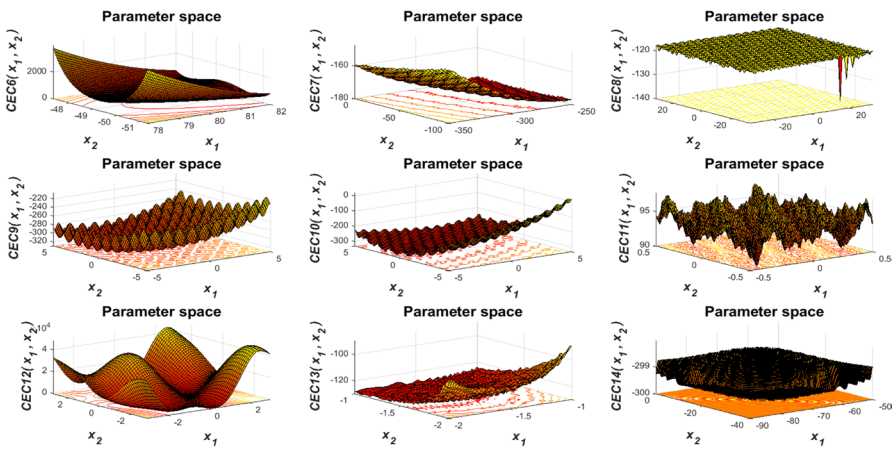
Several research directions can be recommended for future works. Multi-objective, and binary variants of the TDMBBO may be extended to tackle various, multi-objective, discrete, and real-world optimization problems. The development of the migration operator provides neighborhoods search (exploitation) for habitat. Improving the mutation operator can improve the exploration of the algorithm. A few thresholds and selective parameters in the TDMBBO's equations have not been optimally fine-tuned, making it a new path for further work in the future. As well, tackling problems in different fields (i.e., neural networks, deep learning, data mining, scheduling, image processing, industry, smart home, etc.) could be a valuable contribution and beneficial. Also, using multi-objective optimization methods for the p-median problem can certainly be a more appropriate approach.

Appendix

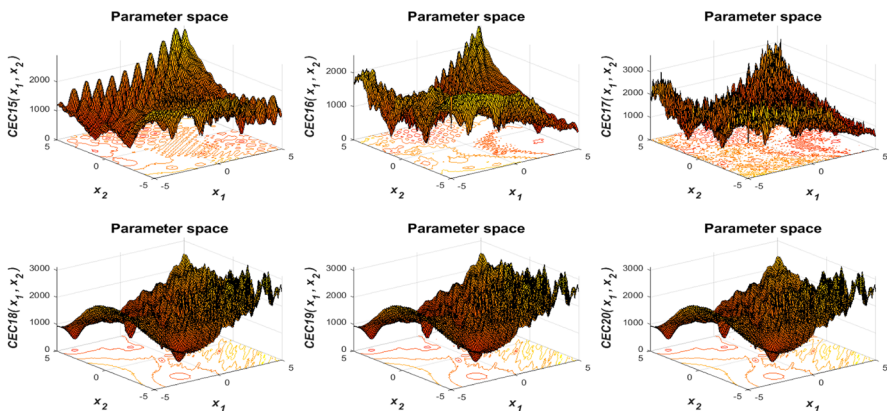
See Figs. 24, 25 and 26.



a) Unimodal CEC2005 test functions



b) Multimodal CEC2005 test functions



c) Composite CEC2005 test functions

Fig. 24 The 3-D plots of CEC2005 benchmark functions

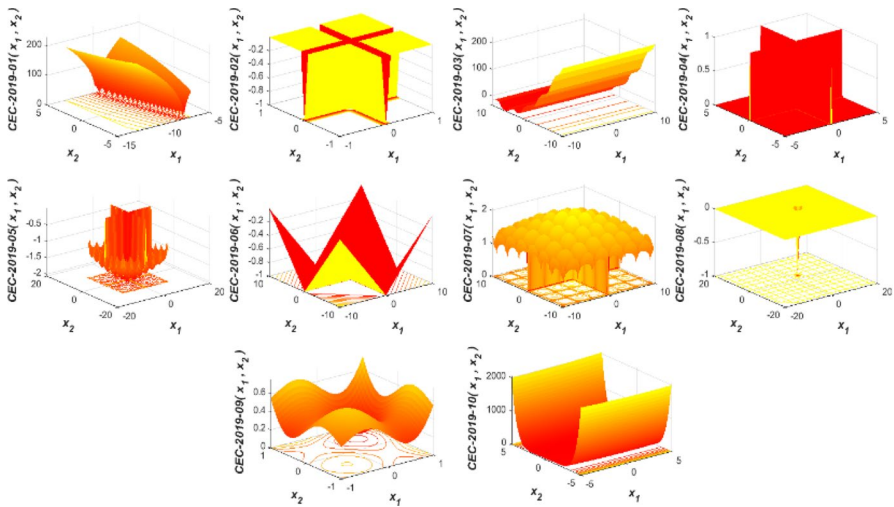


Fig. 25 The 3D visualization of IEEE CEC06-2019 test functions

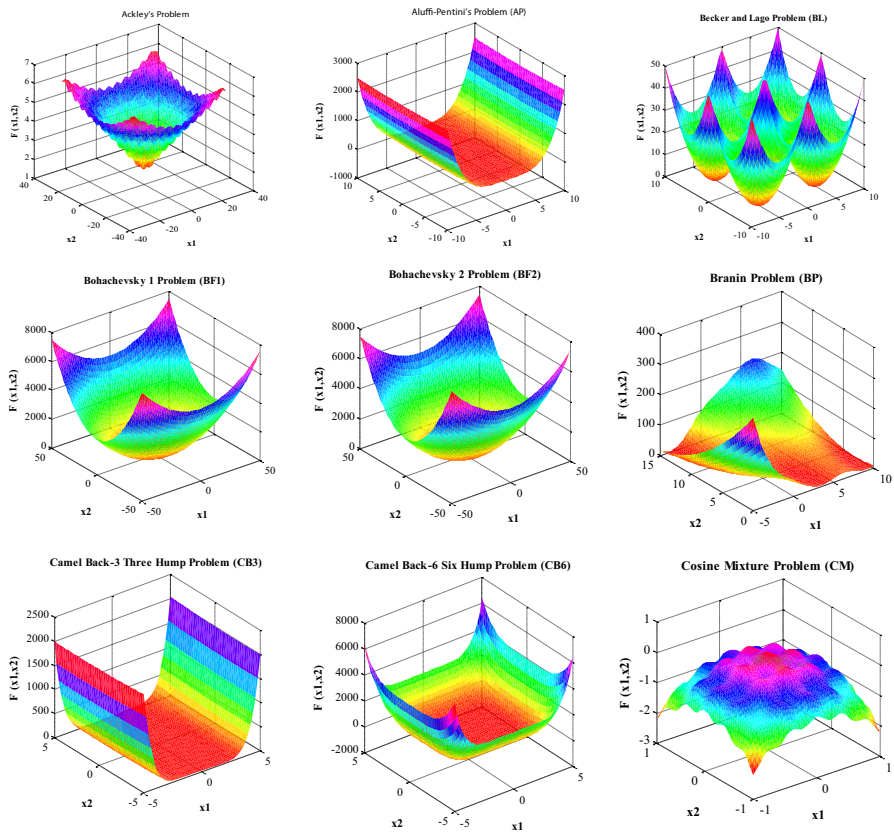


Fig. 26 The 3D visualization of the test functions

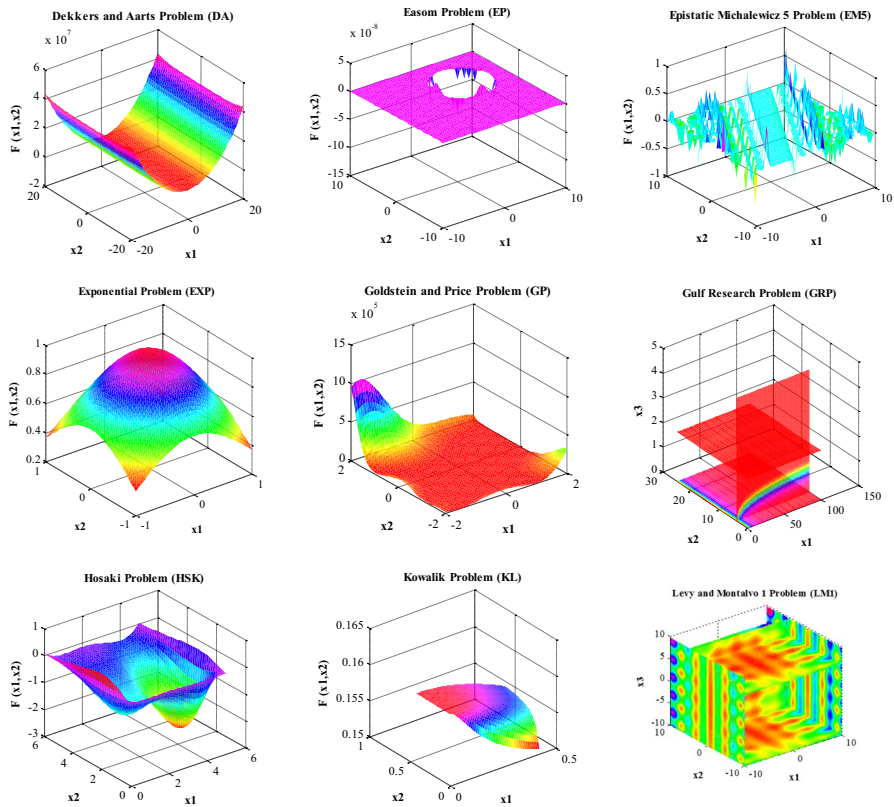


Fig. 26 (continued)

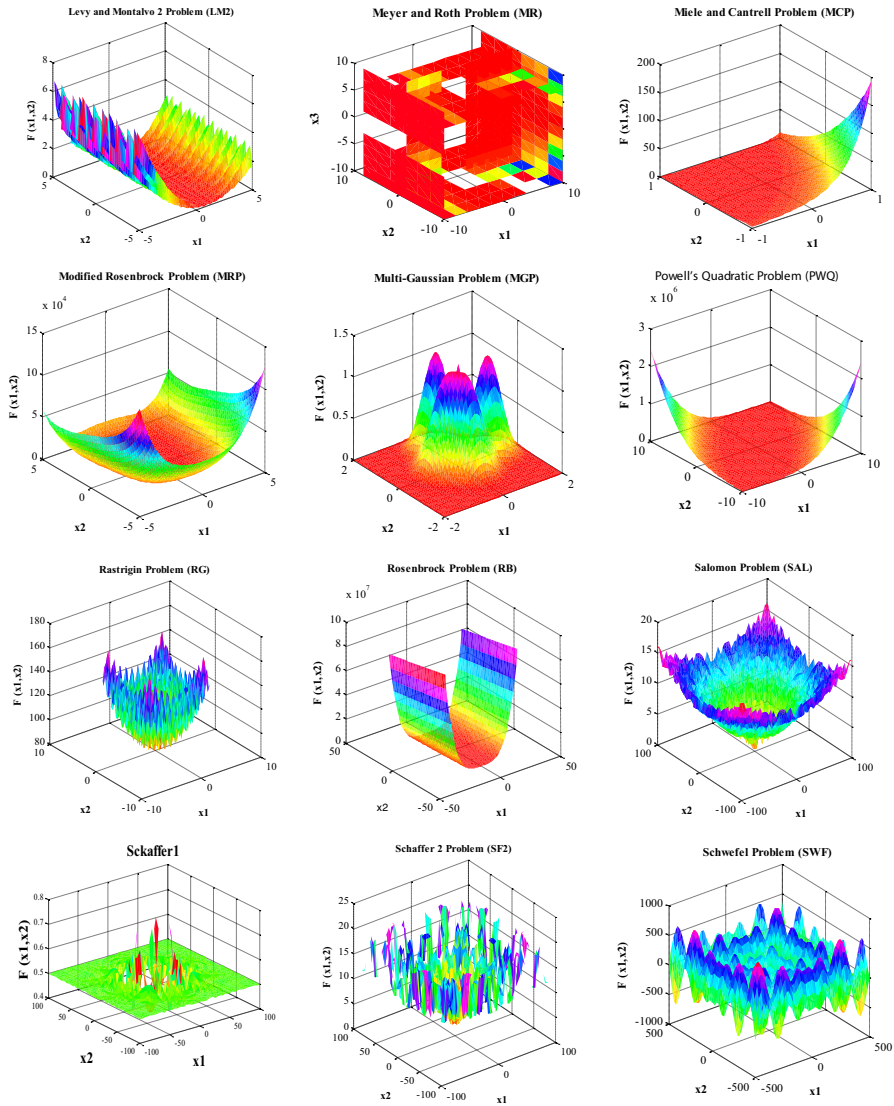


Fig. 26 (continued)

Author contributions This work was carried out in close collaboration among all authors. M. Kaveh conceived the method and experiments, implemented and conducted the experiments. M. S. Mesgari contributed to the data analysis. D. Martín contributed to the experiments and in analyzing the results. M. Kaveh analyzed the results. All authors have contributed to writing the paper.

Declarations

Conflict of interests The authors declare that they have no conflict of interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This paper does not contain any studies with human participants or animals.

Availability of data and material The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

References

1. Zhen L, Wang K, Hu H, Chang D (2014) A simulation optimization framework for ambulance deployment and relocation problems. *Comput Ind Eng* 72:12–23
2. Shariff SR, Moin NH, Omar M (2012) Location allocation modeling for healthcare facility planning in Malaysia. *Comput Ind Eng* 62(4):1000–1010
3. Ko YD, Song BD, Hwang H (2016) Location, capacity and capability design of emergency medical centers with multiple emergency diseases. *Comput Ind Eng* 101:10–20
4. Kaveh M, Mesgari MS (2019) Improved biogeography-based optimization using migration process adjustment: an approach for location-allocation of ambulances. *Comput Ind Eng* 135:800–813
5. Laporte G, Nickel S, Saldanha-da-Gama F (2019) Introduction to location science. Location science. Springer, Cham, pp 1–21
6. Khishe M, Mosavi MR, Kaveh M (2017) Improved migration models of biogeography-based optimization for sonar dataset classification by using neural network. *Appl Acoust* 118:15–29
7. Hakimi SL (1965) Optimum distribution of switching centers in a communication network and some related graph theoretic problems. *Oper Res* 13(3):462–475
8. ReVelle CS, Swain RW (1970) Central facilities location. *Geogr Anal* 2(1):30–42
9. Rolland E, Schilling DA, Current JR (1997) An efficient tabu search procedure for the p-median problem. *Eur J Oper Res* 96(2):329–342
10. Cadenas JM, Canós MJ, Garrido MC, Ivorra C, Liern V (2011) Soft-computing based heuristics for location on networks: the p-median problem. *Appl Soft Comput* 11(2):1540–1547
11. Gokalp O (2020) An iterated greedy algorithm for the obnoxious p-median problem. *Eng Appl Artif Intell* 92:103674
12. Elgüzél İM, Özceylan E (2021) Application of an improved discrete crow search algorithm with local search and elitism on a humanitarian relief case. *Artif Intell Rev* 54(6):4591–4617
13. Kariv, O., & Hakimi, S. L. (1979). An algorithmic approach to network location problems. I: The p-centers. *SIAM Journal on Applied Mathematics*, 37(3), 513–538.
14. Zhang W, Cao K, Liu S, Huang B (2016) A multi-objective optimization approach for health-care facility location-allocation problems in highly developed cities such as Hong Kong. *Comput Environ Urban Syst* 59:220–230
15. ElKady SK, Abdelsalam HM (2016) A modified particle swarm optimization algorithm for solving capacitated maximal covering location problem in healthcare systems. In: *Applications of intelligent optimization in biology and medicine*, pp. 117–133.
16. Zaffar MA, Rajagopalan HK, Saydam C, Mayorga M, Sharer E (2016) Coverage, survivability or response time: a comparative study of performance statistics used in ambulance location models via simulation–optimization. *Oper Res Health Care* 11:1–12
17. Kaidi W, Khishe M, Mohammadi M (2022) Dynamic levy flight chimp optimization. *Knowl-Based Syst* 235:107625
18. Simon D (2008) Biogeography-based optimization. *IEEE Trans Evol Comput* 12(6):702–713
19. Ma H, Simon D (2011) Analysis of migration models of biogeography-based optimization using Markov theory. *Eng Appl Artif Intell* 24(6):1052–1060
20. Shirani MR, Safi-Esfahani F (2021) Dynamic scheduling of tasks in cloud computing applying dragonfly algorithm, biogeography-based optimization algorithm and Mexican hat wavelet. *J Supercomput* 77(2):1214–1272
21. Kaveh M, Kaveh M, Mesgari MS, Paland RS (2020) Multiple criteria decision-making for hospital location-allocation based on improved genetic algorithm. *Appl Geomat* 12(3):291–306
22. Hashim FA, Houssein EH, Hussain K, Mabrouk MS, Al-Atabany W (2022) Honey badger algorithm: new metaheuristic algorithm for solving optimization problems. *Math Comput Simul* 192:84–110

23. Wang J, Khishe M, Kaveh M, Mohammadi H (2021) Binary chimp optimization algorithm (BChOA): a new binary meta-heuristic for solving optimization problems. *Cogn Comput* 13(5):1297–1316
24. Baniasadi S, Rostami O, Martín D, Kaveh M (2022) A novel deep supervised learning-based approach for intrusion detection in IoT systems. *Sensors* 22(12):4459
25. Abdollahzadeh B, Gharehchopogh FS, Mirjalili S (2021) African vultures optimization algorithm: a new nature-inspired metaheuristic algorithm for global optimization problems. *Comput Ind Eng* 158:107408
26. Kaveh M, Mesgari MS (2022) Application of meta-heuristic algorithms for training neural networks and deep learning architectures: a comprehensive review. *Neural Process Lett* 1–104.
27. Goldanloo MJ, Gharehchopogh FS (2022) A hybrid OBL-based firefly algorithm with symbiotic organisms search algorithm for solving continuous optimization problems. *J Supercomput* 78(3):3998–4031
28. Rostami O, Kaveh M (2021) Optimal feature selection for SAR image classification using biogeography-based optimization (BBO), artificial bee colony (ABC) and support vector machine (SVM): a combined approach of optimization and machine learning. *Comput Geosci* 25(3):911–930
29. Azizi, M., Talatahari, S., & Gandomi, A. H. (2022). Fire hawk optimizer: a novel metaheuristic algorithm. *Artif Intell Rev*, 1–77.
30. Kianfar N, Mesgari MS, Mollalo A, Kaveh M (2022) Spatio-temporal modeling of COVID-19 prevalence and mortality using artificial neural network algorithms. *Spat Spatio-temporal Epidemiol* 40:100471
31. Sadeghi F, Rostami O, Yi MK, Hwang SO (2023) A deep learning approach for detecting Covid-19 using the chest X-ray images. *Cmc-Comput Mater Continua* 74(1):751–768
32. Giri PK, De SS, Dehuri S (2021) Adaptive neighbourhood for locally and globally tuned biogeography based optimization algorithm. *J King Saud Univ Comput Inf Sci* 33(4):453–467
33. An Y, Chen X, Li Y, Han Y, Zhang J, Shi H (2021) An improved non-dominated sorting biogeography-based optimization algorithm for the (hybrid) multi-objective flexible job-shop scheduling problem. *Appl Soft Comput* 99:106869
34. Zhang X, Wang D, Chen H (2019) Improved biogeography-based optimization algorithm and its application to clustering optimization and medical image segmentation. *IEEE Access* 7:28810–28825
35. Giri PK, De SS, Dehuri S, Cho SB (2021) Biogeography based optimization for mining rules to assess credit risk. *Intell Syst Account Financ Manag* 28(1):35–51
36. Garg V, Deep K (2016) Performance of laplacian biogeography-based optimization algorithm on CEC 2014 continuous optimization benchmarks and camera calibration problem. *Swarm Evol Comput* 27:132–144
37. Ma H (2010) An analysis of the equilibrium of migration models for biogeography-based optimization. *Inf Sci* 180(18):3444–3464
38. Garg V, Deep K (2015) A state-of-the-art review of biogeography-based optimization. In *Proceedings of Fourth International Conference on Soft Computing for Problem Solving*. Springer, New Delhi, pp. 533–549
39. Wolpert DH, Macready WG (1997) No free lunch theorems for optimization. *IEEE Trans Evol Comput* 1(1):67–82
40. Saaty RW (1987) The analytic hierarchy process—what it is and how it is used. *Math Model* 9(3–5):161–176
41. Li J, Wu Y, Fong S, Tallón-Ballesteros AJ, Yang XS, Mohammed S, Wu F (2022) A binary PSO-based ensemble under-sampling model for rebalancing imbalanced training data. *J Supercomput* 78(5):7428–7463
42. Nabi S, Ahmed M (2022) PSO-RDAL: Particle swarm optimization-based resource-and deadline-aware dynamic load balancer for deadline constrained cloud tasks. *J Supercomput* 78(4):4624–4654
43. Ibrahim S, Jarbouli B (2020) A general variable neighborhood search approach based on a p-median model for cellular manufacturing problems. *Optim Lett* 1–15.
44. Braik M, Sheta A, Al-Hiary H (2021) A novel meta-heuristic search algorithm for solving optimization problems: capuchin search algorithm. *Neural Comput Appl* 33(7):2515–2547
45. Khishe M, Mosavi MR (2020) Chimp optimization algorithm. *Expert Syst Appl* 149:113338
46. Mirjalili S, Mirjalili SM, Lewis A (2014) Grey wolf optimizer. *Adv Eng Softw* 69:46–61

47. Derrac J, García S, Molina D, Herrera F (2011) A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm Evol Comput* 1(1):3–18
48. Suganthan PN, Hansen N, Liang JJ, Deb K, Chen YP, Auger A, Tiwari S (2005) Problem definitions and evaluation criteria for the CEC 2005 special session on real-parameter optimization. *KanGAL Report*, 2005
49. Molina D, Moreno-García F, Herrera F (2017) Analysis among winners of different IEEE CEC competitions on real-parameters optimization: Is there always improvement?. In: 2017 IEEE Congress on Evolutionary Computation (CEC), 805–812
50. Baykasoğlu A, Akpinar Ş (2017) Weighted Superposition Attraction (WSA): a swarm intelligence algorithm for optimization problems—part 1: unconstrained optimization. *Appl Soft Comput* 56:520–540
51. Chetty S, Adewumi AO (2013) Three new stochastic local search algorithms for continuous optimization problems. *Comput Optim Appl* 56(3):675–721

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Mehrdad Kaveh¹ · Mohammad Saadi Mesgari¹ · Diego Martín² · Masoud Kaveh³

✉ Mehrdad Kaveh
m.kaveh11@email.kntu.ac.ir

✉ Mohammad Saadi Mesgari
mesgari@kntu.ac.ir

Diego Martín
diego.martin.de.andres@upm.es

Masoud Kaveh
masoud.kaveh@aalto.fi

¹ Department of Geodesy and Geomatics, K. N. Toosi University of Technology, 19967-15433 Tehran, Iran

² ETSI de Telecomunicación, Universidad Politécnica de Madrid, Av. Complutense 30, 28040 Madrid, Spain

³ Department of Communication and Networking, Aalto University, Helsinki, Finland